VOLUMETRIC ANALYSIS

Class XII Chemistry (NEB), Chapter 1: study notes, solved numericals and past questions

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Tags used in these notes

NEB PYQ year Set Q a question from the NEB board papers, worked solution given.
SXC PYQ *paper Q* a question from the St. Xavier's college papers (1T/2T = first/second terminal, PB = pre-board), solved here.
VERY IMPORTANT asked repeatedly or marked as most important.
TRICK a memory trick or shortcut.
TEXTBOOK section Garuda Basic Chemistry Vol. II, cited by section number.
MANUAL year Set Q NEB solution manual (Asmita), cited by exam year, set and question.

PART 1 Introduction to volumetric analysis

What is quantitative analysis?

Quantitative analysis (Textbook, Introduction)

Quantitative analysis measures the quantity of particular constituents present in a substance. In chemistry it is done by **volumetric**, **gravimetric** and **instrumental** methods.

Volumetric analysis (§1.1.1): quantitative analysis in which the quantity of a substance is found using a known volume of a standard solution or reagent.

Gravimetric analysis (§1.1.2): the quantity of an analyte is found by measuring a change in mass. Four types: physical gravimetry, thermogravimetry, precipitative gravimetric analysis, electrodeposition.

Applications of titration (§1.7.14): checking the concentration of ingredients to ensure quality of soap, pharmaceuticals, other chemical industries and biodiesel; also gives information about impurities. NEB asked: titration in the pharmaceutical industry, e.g. aspirin or vitamin C content, purity of additives.

Chapter map and where the marks are

Part	Topic	Typical exam use
2-4	Equivalent weight, n-factor, $KMnO_4$	MCQ (1 mark), reasoning in long questions
5-8	Concentration terms, conversions, ppm	Numerical 2-5 marks, MCQ
9	Normality equation, dilution, mixing, purity	Derivation (2 marks) + numericals every year
10-11	Back titration, water of crystallisation, gas purity	5 mark numerical, near-certain
12-14	Titration terms, redox titration, f, standards	Definitions, differences, 8 mark long question

Repeat radar: what keeps coming back

Middle column = NEB board years (Manual). Right column = your college papers and the NEB 2082 / 2083 papers.

Question type	NEB Manual years	SXC / NEB-2082, 2083
Deduce $N_1V_1 = N_2V_2$	2081 P/Q, 2080 G/H	1T-2024 Q12, NEB-2082 Q20, PB-2026 Q12
End point vs equivalence point	2080 GIE B, 2081 R, 2075 GIE, 2069 A, 2064	concept, see Part 12
Titration error / normality factor	2080 GIE A, 2076 C, 2074 A, 2074 Supp, 2073	1T-2024 Q21, 2T-2024 Q12, 2T-2025 Q12
Primary vs secondary standard	2080 J, 2076 GIE A/B, 2075 B, 2071 C, 2073 D	1T-2024 Q21, PB-2025 Q20, NEB-2083 Q12
$KMnO_4$ equivalent weight; why dil. H_2SO_4	2079 D, 2079 H, 2081 P/Q	PB-2025 Q20, NEB-2082 Q20
Add water to dilute (to N/10 etc.)	2077 V, 2080 GIE A, 2079 D, 2070 D, 2069 A	1T-2025 Q6, 2T-2024 Q12
Mix two solutions for a target	2067, 2071 C, 2074 Supp, 2073 C/D	PB-2025 Q20, PB-2026 Q12
Divalent metal in excess acid	2078 D, 2076 B/C, 2076 GIE B, 2074 A, 2072 C, 2070 Supp, 2065, 2079 D	2T-2025 Q12
$Na_2CO_3.xH_2O$ find x	2079 GIE B, 2081 P/Q, 2064	1T-2024 Q12
% purity (limestone, zinc, H_2SO_4)	2075 GIE (Zn), 2072 E (H_2SO_4)	1T-2025 Q15
Conc. acid (% w/w, density): N, M, molality, volume	2080 GIE B, 2077 W, 2072 Supp, 2075 B	2T-2024 Q12, 1T-2025 Q13
N from M (n-factor), MCQ	2081 P/Q, 2081 R, 2080 J, 2079 GIE A	PB-2025 Q1, NEB-2083 Q2

PART 2 Equivalent weight: general formula, n-factor of acid

Equivalent weight and the n-factor

General formula

The formula is what numericals need. The n-factor is also called the valency factor or z-factor.

$$E = \frac{\text{Atomic weight}}{n} \text{ (atoms)} \qquad E = \frac{\text{Molecular weight}}{n} \text{ (molecules)} \qquad E = \frac{\text{Ionic weight}}{n} \text{ (ions)}$$

Substance	n-factor is
Acid	basicity = number of REPLACEABLE H^+ per molecule
Base	acidity = number of REPLACEABLE OH^- per molecule
Salt	total positive charge (= total negative charge); use either, do not add
Oxidising / reducing agent	change in oxidation number per formula unit

Why "basicity" for an acid: one mole of acid neutralises that many moles of base. A replaceable H is one that can leave as H^+, i.e. H bonded to a very electronegative atom (Cl, or the O of an -OH).

Textbook: equivalent mass of an element and of an acid

Equivalent mass of an element (§1.1.3, §1.2)

The number of parts by weight of the element that combine with or displace, directly or indirectly, **1.008 parts of hydrogen, or 8 parts of oxygen, or 35.5 parts of chlorine**. Example: E of Al = 27/3 = 9.

Derivation of E = A/V (§1.2): let the element have atomic mass A and valency V.

1. Valency V means one atom of the element combines with V atoms of hydrogen.
2. So V × 1.008 parts by mass of hydrogen combine with A parts by mass of the element.
3. Hence 1.008 parts of hydrogen combine with A/V parts of the element.
4. By definition that mass is the equivalent mass.

$$E = \frac{\text{Atomic mass (A)}}{\text{Valency (V)}}$$

Equivalent mass of an acid (§1.3.1)

The number of parts by weight of acid that contain **1.008 parts by weight of replaceable hydrogen**.

$$E_{acid} = \frac{\text{Molecular mass}}{\text{Basicity}}$$

Basicity = number of replaceable (ionisable) hydrogen atoms per molecule.

Acid	Formula	Basicity	Equivalent mass
Hydrochloric	HCl	1	36.5
Sulphuric	H_2SO_4	2	98/2 = 49
Nitric	HNO_3	1	63
Acetic	CH_3COOH	1	60

Acid	Formula	Basicity	Equivalent mass
Formic	HCOOH	1	46
Oxalic (anhydrous)	$(COOH)_2$	2	90/2 = 45
Oxalic crystal	$(COOH)_2.2H_2O$	2	126/2 = 63
Phosphoric	H_3PO_4	3	98/3 = 32.67
Phosphorous	H_3PO_3	2	82/2 = 41
Hypophosphorous	H_3PO_2	1	66

Textbook table in §1.3.1, extended with H_3PO_4 (3 + 31 + 64 = 98).

Acids of phosphorus (frequently asked)

VERY IMPORTANT **H_3PO_4, H_3PO_3, H_3PO_2: n-factor 3, 2, 1**

Acid	Name	H on O	H on P	n-factor
H_3PO_4	phosphoric (ortho)	3	0	3
H_3PO_3	phosphorous	2	1	2
H_3PO_2	hypophosphorous ("hypo" = less oxygen)	1	2	1

Only H bonded to O is replaceable. H bonded directly to P is NOT replaceable.

TRICK **n = (number of O atoms) − 1**, valid only for these three acids: H_3PO_4: 4−1 = 3; H_3PO_3: 3−1 = 2; H_3PO_2: 2−1 = 1. Do not use it for any other acid.

Class question: n-factor of (A) H_3PO_4 (B) H_3PO_3 (C) H_3PO_2 = **3, 2, 1**.

Solved examples

Example 2.1 **Q.** Find the equivalent weight of HCl.

Molecular weight = 1 + 35.5 = 36.5; HCl has 1 replaceable H, so n = 1.

$$E = \frac{36.5}{1} = 36.5$$

Answer: 36.5 g/eq

Example 2.2 **Q.** Find the equivalent weight of H_2SO_4.

Molecular weight = 2(1) + 32 + 4(16) = 98; both H are replaceable, so n = 2.

$$E = \frac{98}{2} = 49$$

Answer: 49 g/eq

TEXTBOOK §1.3.1 table same value, 49.

Textbook Solved Question 1 (H_3PO_4 in three reactions) **TEXTBOOK** §1.3.1 Solved Q1 **VERY IMPORTANT** **Q.** Calculate the equivalent mass of H_3PO_4 in the reactions: (i) $NaOH + H_3PO_4 \rightarrow NaH_2PO_4 + H_2O$; (ii) $2NaOH + H_3PO_4 \rightarrow Na_2HPO_4 + 2H_2O$; (iii) $3NaOH + H_3PO_4 \rightarrow Na_3PO_4 + 3H_2O$.

n-factor = number of H atoms actually replaced by Na in that reaction. Molecular mass of H_3PO_4 = 3 + 31 + 64 = 98.

(i) one H replaced, n = 1: E = 98/1 = 98.

(ii) two H replaced, n = 2: E = 98/2 = 49.

(iii) three H replaced, n = 3: E = 98/3 = 32.67.

Answer: (i) 98 (ii) 49 (iii) 32.67

NEB PYQ 2081 Set R Q1 part (i) as an MCQ: options 2, 49, 59, 98, answer 98 (option d).

Textbook "Test your knowledge" **TEXTBOOK** §1.6 end **Q.** If the equivalent mass of metal M is 9, what is the molecular mass of M_2O_3?

Valency of M in M_2O_3 is 3, so atomic mass of M = E × valency = 9 × 3 = 27.

Molecular mass of M_2O_3 = 2(27) + 3(16) = 54 + 48 = 102.

Answer: 102

PYQ drill: equivalent weight and n-factor

NEB PYQ 2079 GIE Set B Q10 **Equivalent weight of a metal from a chloride ratio** **Q.** If the weight of metal and chlorine in a metal chloride is 1 : 2, the equivalent weight of the metal is (a) 71 (b) 35.5 (c) 17.75 (d) 8.875.

The metal combines with chlorine in proportion to equivalent weights, and E(Cl) = 35.5.

$$\frac{E}{35.5} = \frac{1}{2} \Rightarrow E = \frac{35.5}{2} = 17.75$$

Answer: (c) 17.75

NEB PYQ 2079 GIE Set A Q1 **n-factor of H_3PO_3 used in a normality MCQ** **Q.** What is the normality of 0.3 M phosphorous acid (H_3PO_3)? (a) 0.1 (b) 0.2 (c) 0.6 (d) 0.9

H_3PO_3 is dibasic (2 replaceable H), so n = 2.

N = M × n = 0.3 × 2 = 0.6 N.

Answer: 0.6 N, option (c). The manual's printed key says (d) but its own working gives 0.6, which is option (c).

SXC PYQ *1T-2024 Q1* **Equivalent mass of barium from a precipitation** **Q.** 2.1 g of anhydrous $BaCl_2$ is completely converted into 2.36 g of $BaSO_4$ by excess H_2SO_4. Equivalent mass of barium (Ba = 137) is (a) 25.45 (b) 65.46 (c) 72.6 (d) 42.68.

Law of equivalence: g-eq of $BaCl_2$ = g-eq of $BaSO_4$. Let E = equivalent mass of Ba; E(Cl) = 35.5, E(SO_4) = 48.

$$\frac{2.1}{E + 35.5} = \frac{2.36}{E + 48}$$

$2.1(E + 48) = 2.36(E + 35.5) \Rightarrow 2.1E + 100.8 = 2.36E + 83.78$

$17.02 = 0.26\ E \Rightarrow E = 17.02/0.26 = 65.46$

Answer: 65.46, option (b)

PART 3 Equivalent weight of base and salt (ionisation method)

Base and salt

Key points and tricks

Base: E = molecular weight ÷ acidity (number of replaceable OH^-).

Salt: ionise first, then E = molecular weight ÷ n, where n = total positive charge = total negative charge (use one, never add them). General idea $A_xB_y \rightarrow x\ A^{y+} + y\ B^{x-}$, mostly valid, not for all salts.

Textbook definitions (§1.3.2, §1.3.3, §1.3.5)

Equivalent mass of a base: the number of parts by weight of base completely neutralised by one gram equivalent of an acid. Acidity = number of ionisable OH^- in a molecule.

$$E_{base} = \frac{\text{Molecular mass}}{\text{Acidity}}$$

Equivalent mass of a salt: the number of parts by weight of salt produced by neutralising one gram equivalent of an acid by a base (or vice versa).

$$E_{salt} = \frac{\text{Molecular mass}}{\text{No. of +ve / −ve charge}}$$

Equivalent mass of a radical or ion:

$$E_{radical} = \frac{\text{Ionic mass}}{\text{No. of charge on the radical}}$$

Salt	Ionisation	n	Equivalent mass
NaCl	$NaCl \rightarrow Na^+ + Cl^-$	1	58.5
KCl	$KCl \rightarrow K^+ + Cl^-$	1	74.5
$CaCO_3$	$CaCO_3 \rightleftharpoons Ca^{2+} + CO_3^{2-}$	2	100/2 = 50
Na_2CO_3	$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-}$	2	106/2 = 53
Na_2SO_4	$Na_2SO_4 \rightarrow 2Na^+ + SO_4^{2-}$	2	142/2 = 71
$Al_2(SO_4)_3$	$Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$	6	342/6 = 57

Base	Acidity	Equivalent mass
NaOH	1	23 + 16 + 1 = 40
KOH	1	39 + 16 + 1 = 56
$Mg(OH)_2$	2	(24 + 34)/2 = 29
$Ca(OH)_2$	2	(40 + 34)/2 = 37

Textbook §1.3.2 table.

Solved examples

Example 3.1 **Q.** Equivalent weight of NaOH.

M = 23 + 16 + 1 = 40; acidity = 1.

$$E = \frac{40}{1} = 40$$

Answer: 40

TEXTBOOK §1.3.2 NaOH = 40.

Example 3.2 **Q.** Equivalent weight of $Mg(OH)_2$.

M = 24 + 2(17) = 58; acidity = 2.

$$E = \frac{58}{2} = 29$$

Answer: 29

Example 3.3 **Q.** Equivalent weight of Na_2CO_3.

$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-}$. Total positive charge = 2 × 1 = 2; total negative charge = 2. So n = 2.

M = 2(23) + 12 + 3(16) = 106.

$$E = \frac{106}{2} = 53$$

Answer: 53

TEXTBOOK §1.3.3 table Na_2CO_3 = 53, the value used in most numericals.

Example 3.4 **Q.** n-factor of $Al_2(SO_4)_3$ (salt of $Al(OH)_3$ and H_2SO_4).

$Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$

Positive charge = 2 × 3 = 6; negative charge = 3 × 2 = 6. n = 6, E = 342/6 = 57.

Answer: n = 6

SXC PYQ *PB-2025 Q1* normality of 0.1 M $Al_2(SO_4)_3$ = 0.1 × 6 = 0.6 N, option (D), see Part 6.

Textbook examples of radicals and hydrated salts TEXTBOOK §1.3.5, Solved Numerical 1 **Q.** Equivalent masses of NO_3^-, SO_4^{2-} and washing soda $Na_2CO_3.10H_2O$.

$$E(NO_3^-) = \frac{14 + 16\times3}{1} = 62 \qquad E(SO_4^{2-}) = \frac{32 + 16\times4}{2} = 48$$

$$E(Na_2CO_3.10H_2O) = \frac{2(23) + 12 + 48 + 10(18)}{2} = \frac{286}{2} = 143$$

Water of crystallisation raises the molecular mass but does not change n (n comes from the Na_2CO_3 part only).

Answer: 62, 48, 143

Equivalent weights to memorise

Substance	E	Substance	E	Substance	E
HCl	36.5	NaOH	40	Na_2CO_3	53
H_2SO_4	49	KOH	56	$CaCO_3$	50
HNO_3	63	$Ca(OH)_2$	37	$NaHCO_3$	84
Oxalic crystal	63	$Mg(OH)_2$	29	$Na_2CO_3.10H_2O$	143
Oxalic anhydrous	45	NH_4Cl	53.5	$FeSO_4.7H_2O$	278
CH_3COOH	60	$KMnO_4$ (acid)	31.6	$K_2Cr_2O_7$ (acid)	49

Computed from H 1, C 12, N 14, O 16, Na 23, Mg 24, S 32, Cl 35.5, K 39, Ca 40, Mn 55, Fe 56, Cr 52.

PART 4 $KMnO_4$ as oxidising agent: n-factor and equivalent weight in acidic, basic, neutral media

Oxidising agents and the n-factor from a reaction

Key points and tricks

For an oxidising or reducing agent, **n = change in oxidation number** per formula unit. Oxidising agent: oxidises others, is itself reduced (oxidation number falls). Reducing agent: reduces others, is itself oxidised.

Oxidation numbers worked out: $KMnO_4$: +1 + x − 8 = 0, so Mn = +7. $MnSO_4$: +2. K_2MnO_4: 2 + x − 8 = 0, so +6. MnO_2: +4.

VERY IMPORTANT **Equivalent weight of $KMnO_4$ in three media (M = 39 + 55 + 64 = 158)**

Medium	Added	Reaction (no balancing needed)	Mn change	n	E
Acidic	dil. H_2SO_4	$KMnO_4 + H_2SO_4 \rightarrow K_2SO_4 + MnSO_4 + H_2O + [O]$	+7 → +2	5	158/5 = 31.6
Basic	KOH	$KMnO_4 + KOH \rightarrow K_2MnO_4 + H_2O + [O]$	+7 → +6	1	158/1 = 158
Neutral	H_2O	$KMnO_4 + H_2O \rightarrow KOH + MnO_2 + [O]$	+7 → +4	3	158/3 = 52.67

TRICK **"B.A.N. = Bangladesh 153"** (memory joke): Basic = 1, Acidic = 5, Neutral = 3, so 1 5 3. Never forget acidic = 5.

The reactions must be remembered if the question asks for E in different media (products only; balancing not needed). [O] is nascent oxygen. Potassium dichromate is also an oxidising agent (acidic medium).

Textbook balanced forms (§1.3.4):

$$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$$

$$2KMnO_4 + 2KOH \rightarrow 2K_2MnO_4 + H_2O + [O] \qquad 2KMnO_4 + H_2O \rightarrow 2KOH + 2MnO_2 + 3[O]$$

Textbook: equivalent mass of $K_2Cr_2O_7$ (oxidising agent, §1.3.4)

$$K_2Cr_2O_7 + 4H_2SO_4 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3[O]$$

Cr changes +6 → +3 in each of 2 Cr atoms: total change = 2(6 − 3) = 6.

$$E = \frac{2(39) + 2(52) + 7(16)}{6} = \frac{294}{6} = 49$$

Past question: n-factor depends on the reaction

Example 4.1 (past question) **VERY IMPORTANT** TEXTBOOK §1.3.1 Solved Q1 (ii) **Q.** What is the equivalent weight of H_3PO_4 in the reaction $NaOH + H_3PO_4 \rightarrow Na_2HPO_4 + H_2O$?

Balance: $H_3PO_4 + 2NaOH \rightarrow Na_2HPO_4 + 2H_2O$.

Key idea: n depends on how many H^+ are actually replaced in THIS reaction, not on how many H the acid has. One H stays in Na_2HPO_4, so only 2 of 3 are replaced: n = 2.

$$E = \frac{98}{2} = 49$$

If all three H were replaced (Na_3PO_4) n would be 3. Always look at the product.

Answer: 49

Example 4.2 (oxidation numbers) **Q.** Find the oxidation number of C in oxalic acid $H_2C_2O_4$ and of Mn in $KMnO_4$.

$H_2C_2O_4$: 2(+1) + 2x + 4(−2) = 0 ⇒ 2x = 6 ⇒ x = +3.

$KMnO_4$: (+1) + x + 4(−2) = 0 ⇒ x = +7.

In the $KMnO_4$ / oxalic titration C goes +3 → +4 (2 C atoms, n = 2) and Mn goes +7 → +2 (n = 5). This is why 2 $KMnO_4$ react with 5 $H_2C_2O_4$.

Answer: C = +3, Mn = +7

PYQ drill: n-factor of oxidising agents

SXC PYQ *PB-2026 Q1* **Equivalent weight of $K_2Cr_2O_7$ when no oxidation number changes Q.** The equivalent mass of $K_2Cr_2O_7$ in the reaction $K_2Cr_2O_7 + 2KOH \rightarrow 2K_2CrO_4 + H_2O$ is (a) 294 (b) 147 (c) 98 (d) 49.

Cr stays +6 in both $K_2Cr_2O_7$ and K_2CrO_4, so this is not a redox reaction and the 49 of the acidic redox reaction does NOT apply. One mole of $K_2Cr_2O_7$ reacts with 2 moles of KOH, so n = 2.

$$E = \frac{294}{2} = 147$$

Answer: 147, option (b)

NEB PYQ 2079 Set D Q21 ***SXC PYQ*** *PB-2025 Q20* **NEB PYQ** 2082 Set D Q20 **Equivalent weight of $KMnO_4$ in acidic medium Q.** Calculate the equivalent mass of $KMnO_4$ in acidic medium (molar mass 158).

$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5[O]$. Change in oxidation number of Mn = (+7) − (+2) = 5.

$$E = \frac{\text{Molecular weight}}{\text{Change of O.N.}} = \frac{158}{5} = 31.6$$

Answer: 31.6

PART 5 Concentration terms: % w/v, % w/w, g/L, normality, molarity, molality

Concentration terms

Key points and tricks

Concentration = amount of SOLUTE in a SOLUTION (molality is the one exception: it uses the SOLVENT). Volume of solution is in **litres** everywhere except % w/v (mL). Weights are in grams; molality's solvent weight is in **kg**.

Term	Formula	Units to use
% w/v	(g solute / mL solution) × 100	g and mL
% w/w	(g solute / g solution) × 100	g and g (solution = solute + solvent)
g/L	g solute / L solution	g and L
Normality N	g-eq solute / L solution; g-eq = W/E	W in g, V in L
Molarity M	moles solute / L solution; moles = W/M_w	W in g, V in L
Molality m	moles solute / kg SOLVENT	solvent in kg

TRICK "W/M = washing machine": moles = W/M_w. Capital M = molarity (solution); small m = molality (solvent). Never mix them up.

20 % w/v NaOH = 20 g NaOH in 100 mL solution. **20 % w/w urea** = 20 g urea in 100 g solution (80 g solvent). If only "%" is given, take it as % w/v.

Names: N = normal (1), N/2 = semi-normal (0.5), N/5 = penti-normal (0.2), N/10 = deci-normal (0.1), N/100 = centi-normal (0.01). Same words with M for molar and m for molal. Molecular weight is never given: work it out.

$$W = \frac{N \times E \times V(mL)}{1000}$$

Textbook: concentration of solution (§1.4)

Percentage and gram per litre (§1.4.1, §1.4.2)

% w/w: mass of solute in grams present in 100 g of solution. 10 % H_2SO_4 by weight = 10 g in 100 g solution.
% w/V: mass of solute in grams present in 100 mL of solution. 10 % H_2SO_4 by volume = 10 g in 100 mL solution.
% V/V: volume of solute in mL present in 100 mL of solution. 40 % ethyl alcohol V/V = 40 mL alcohol in 100 mL solution.
Gram per litre: mass in grams dissolved in one litre of solution. 20 g/L Na_2CO_3 = 20 g in 1 L.

Molarity (§1.4.3)

The number of moles of solute dissolved in one litre of solution. Symbol M; unit mol/L.

$$M = \frac{\text{No. of moles of solute}}{\text{Volume of solution (L)}} = \frac{w}{M_w \times V(L)} = \frac{w \times 1000}{M_w \times V(mL)}$$

Name	Symbol	Moles / L	Example with H_2SO_4 (98 g/mol)
Molar	M	1	98 g in 1 L
Semi-molar	M/2	0.5	49 g in 1 L
Deci-molar	M/10	0.1	9.8 g in 1 L
Centi-molar	M/100	0.01	0.98 g in 1 L
Milli-molar	M/1000	0.001	0.098 g in 1 L

Molality (§1.4.4)

The number of moles of solute present in one kilogram of SOLVENT. Symbol m.

$$m = \frac{\text{No. of moles of solute}}{\text{Mass of solvent (kg)}} = \frac{w}{M_w} \times \frac{1000}{\text{Mass of solvent (g)}}$$

Normality (§1.4.5)

The number of gram equivalents of solute present in one litre of solution. Symbol N; unit g-eq/L.

$$N = \frac{\text{No. of g-eq of solute}}{\text{Volume of solution (L)}} = \frac{W}{E \times V(L)} = \frac{W \times 1000}{E \times V(mL)}$$

Name	Symbol	g-eq / L	Example with HCl (E = 36.5)
Normal	N	1	36.5 g in 1 L
Semi-normal	N/2	0.5	36.5/2 g in 1 L
Deci-normal	N/10	0.1	3.65 g in 1 L
Centi-normal	N/100	0.01	0.365 g in 1 L
Milli-normal	N/1000	0.001	0.0365 g in 1 L

Formality (§1.4.6) is used for ionic compounds in place of molarity: gram formula masses of solute per litre of solution.

For competitive exams (Textbook)

Molality and mole fraction are preferred to molarity and normality because they use WEIGHTS, which do not change with temperature, whereas volumes of solution do. So N and M vary with temperature; molality and mole fraction do not. This is the answer to "why is molality better than molarity?"

Solved examples

Textbook Solved Question 1 (molality) TEXTBOOK §1.4 Solved Q1 **Q.** Calculate the molality of a solution having 9.8 g of H_2SO_4 dissolved in 200 g of water.

$$m = \frac{9.8}{98} \times \frac{1000}{200} = 0.1 \times 5 = 0.5$$

Answer: 0.5 m

Textbook Solved Question 2 (normality) TEXTBOOK §1.4 Solved Q2 **Q.** Calculate the normality of a solution containing 5.3 g of anhydrous Na_2CO_3 in 500 mL of solution.

E = 106/2 = 53.

$$N = \frac{5.3 \times 1000}{53 \times 500} = 0.2$$

Answer: 0.2 N

Alternative route: find M = 0.1 first, then N = 0.1 × 2 = 0.2. Both routes agree.

Textbook Solved Question 3 (molarity) TEXTBOOK §1.4 Solved Q3 **Q.** Calculate the molarity of a solution having 6.3 g of oxalic acid crystal $(COOH)_2.2H_2O$ dissolved in 750 mL of solution.

$M_w = 126$.

$$M = \frac{6.3 \times 1000}{126 \times 750} = 0.0667$$

Answer: 0.0667 M (0.0667 = 1/15)

Normality of NaOH solution NEB PYQ 2062 Q13 **Q.** What is the normality of 500 mL of sodium hydroxide solution containing 30 g NaOH?

g-eq = 30/40 = 0.75; V = 0.5 L.

$$N = \frac{0.75}{0.5} = 1.5$$

Answer: 1.5 N

TEXTBOOK §1.7 Solved Numerical 2 the same question.

Normality and molarity of 5 % NaOH NEB PYQ 2071 Supp Q2 **Q.** Calculate the normality and molarity of 5 % NaOH solution.

5 % means 5 g in 100 mL (w/v). g/L = 50.

$$N = \frac{g/L}{E} = \frac{50}{40} = 1.25$$

For NaOH, $E = M_w = 40$ (n = 1), so M = N.

Answer: N = M = 1.25

g/L and normality of 2 M H_2SO_4 NEB PYQ 2072 Supp Q2 **Q.** Convert 500 cc of 2 M H_2SO_4 into (i) g/L (ii) normality.

(i) g/L = M × M_w = 2 × 98 = 196.

$$\text{(ii) } N = \frac{g/L}{E} = \frac{196}{49} = 4$$

The volume 500 cc does not affect either answer.

Answer: 196 g/L; 4 N

PYQ drill: definitions and differences (write these in the exam)

NEB PYQ 2076 Set B Q2 **Distinguish between decinormal and decimolar solution** **Q.** Distinguish between decinormal and decimolar solution. [2]

Decinormal	Decimolar
One-tenth g-equivalent of solute in 1 L of solution	One-tenth mole of solute in 1 L of solution
Denoted N/10	Denoted M/10
4 g NaOH in 1 L (E = 40)	9.8 g H_2SO_4 in 1 L (M_w = 98)

NEB PYQ 2075 Set A Q2 **NEB PYQ** 2080 Set G/H Q20 **Distinguish between normality and molarity; define each Q.** Distinguish between normality and molarity. [2]

Normality	Molarity
No. of gram equivalents of solute per litre of solution	No. of moles of solute per litre of solution
Denoted N; N = W/(E × V in L)	Denoted M; M = W/(M_w × V in L)
Depends on the reaction (n-factor changes)	Independent of the reaction

NEB PYQ 2076 Set C Q23 ***SXC PYQ*** *2T-2024 Q13a* **Why molality is better than molarity Q.** Why is molality considered better than molarity for expressing concentration? [1]

Molarity uses volume of solution, which expands or contracts with temperature, so molarity changes with temperature. Molality uses the mass of solvent, which does not change with temperature, so molality is constant.

SXC PYQ *2T-2024 Q13b* **What does a 2 M HCl label mean? Q.** A solution of HCl is labelled 2 M. What does it mean? [1]

2 moles of HCl (2 × 36.5 = 73 g) are present in every 1 litre of the solution.

SXC PYQ *1T-2024 Q21d* **Molar and normal solutions Q.** What are molar and normal solutions? [1]

Molar solution: 1 mole of solute dissolved in 1 L of solution (e.g. 98 g H_2SO_4 per litre). **Normal solution:** 1 gram-equivalent of solute in 1 L of solution (e.g. 36.5 g HCl or 49 g H_2SO_4 per litre).

NEB PYQ 2070 Supp Q2 **Semi-normal solution; define Q.** Define semi-normal solution. [1]

A solution containing half a gram-equivalent of solute in one litre of solution; denoted N/2 or 0.5 N.

PART 6 Relation between normality and molarity; N-terms and M-terms; mole fraction

Relation between N and M; mole fraction

VERY IMPORTANT **N = M × n-factor ("very, very important formula")**

Why: g/L = N × E and g/L = M × M_w. So N × E = M × M_w, and since E = M_w/n,

$$N = M \times n$$

For an acid n = basicity; for a base n = acidity; for a salt n = total charge. Normality is always ≥ molarity. The volume given in the question (e.g. 20 cc) is irrelevant to concentration.

M-terms are the same words as N-terms: M molar, M/2 semi-molar, M/10 deci-molar, M/100 centi-molar. The words deci-molal, centi-molal apply to molality.

Textbook: relation between normality and molarity

Derivation (Textbook §1.4.5 and NEB 2065 Q17, 2080 Set J Q12)

Take one litre of solution containing W grams of solute, equivalent mass E and molecular mass M_w, with n = M_w/E.

$$N = \frac{W}{E \times 1} \qquad M = \frac{W}{M_w \times 1}$$

$$\frac{N}{M} = \frac{M_w}{E} = n \quad \Rightarrow \quad N = M \times n$$

Also: N × E = M × M_w = g/L. Hence N = M × basicity (acid) = M × acidity (base).

Substance	n	1 M is	Substance	n	1 M is
HCl, HNO_3, NaOH, KOH	1	1 N	H_2SO_4, $Ca(OH)_2$, Na_2CO_3	2	2 N
H_3PO_3	2	2 N	H_3PO_4, $Al(OH)_3$	3	3 N
Oxalic acid	2	2 N	$Al_2(SO_4)_3$	6	6 N

Mole fraction

$$\chi_{solute} = \frac{n_{solute}}{n_{solute} + n_{solvent}} \qquad \chi_{solvent} = \frac{n_{solvent}}{n_{solute} + n_{solvent}}$$

Sum of all mole fractions in a mixture = 1 exactly (add the fractions: same denominator, numerators add up to it). It is asked as an MCQ: "less than 1 / greater than 1 / 0 / 1", answer 1. Mole fraction has no unit and does not change with temperature.

Solved examples

Normality of 0.4 N to molarity **NEB PYQ** 2081 Set P/Q Q1 **Q.** What is the molarity of 0.4 N Na_2CO_3 solution? (a) 0.1 M (b) 0.2 M (c) 0.4 M (d) 0.8 M

n for Na_2CO_3 = 2.

$$M = \frac{N}{n} = \frac{0.4}{2} = 0.2$$

Answer: (b) 0.2 M

2 M Na_2CO_3 in normality **NEB PYQ** 2080 Set J Q1 **Q.** 2 M Na_2CO_3 is equal to its normality: (a) 1 N (b) 2 N (c) 3 N (d) 4 N

N = M × n = 2 × 2 = 4.

Answer: (d) 4 N

Normality of 0.1 M $Al_2(SO_4)_3$ ***SXC PYQ*** *PB-2025 Q1* **VERY IMPORTANT** **Q.** What is the normality of 0.1 M $Al_2(SO_4)_3$ solution? (A) 0.1 (B) 0.2 (C) 0.3 (D) 0.6

$Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$: n = 6.

N = 0.1 × 6 = 0.6.
Answer: (D) 0.6 N

pH 7 by neutralising H_2SO_4 with NaOH NEB PYQ 2063 Q26 **Q.** 1 g of NaOH is added to 2 L of x M H_2SO_4 so that the pH of the resulting solution is 7. Find x.

pH 7 means exact neutralisation: g-eq of acid = g-eq of NaOH.
g-eq NaOH = 1/40 = 0.025. N of H_2SO_4 = 2x, so g-eq acid = 2x × 2 = 4x.

$$4x = 0.025 \Rightarrow x = \frac{0.025}{4} = 0.00625$$

Answer: $x = 6.25 \times 10^{-3}$ M

Concentration of nitrate ions on mixing NEB PYQ 2079 Set H Q9 **Q.** What is the concentration of nitrate ions if equal volumes of 1 M $NaNO_3$ and 1 M KCl are mixed? (a) 0.25 M (b) 0.5 M (c) 1 M (d) 2 M

Take 1 L of each: 1 mol NO_3^- in 2 L of mixture.

$$[NO_3^-] = \frac{1}{2} = 0.5$$

Answer: (b) 0.5 M

Mole fraction MCQ and one more VERY IMPORTANT **Q.** (i) The sum of all mole fractions of a solution is (less than 1 / greater than 1 / 0 / 1). (ii) A solution has 2 mol of solute and 8 mol of solvent: find χ of each.

(i) 1.
(ii) total = 10; χ(solute) = 2/10 = 0.2; χ(solvent) = 8/10 = 0.8; sum = 1.
Answer: (i) 1 (ii) 0.2 and 0.8

PART 7 ppm and ppb

Ppm and ppb

Key points and tricks

million = 10^6, billion = 10^9. Same as the % formulas but multiplied by 10^6 (ppm) or 10^9 (ppb) instead of 100. Use grams for weights and mL for volumes.

$$\text{ppm} = \frac{\text{W solute (g)}}{\text{W / V of solution (g / mL)}} \times 10^6 \qquad \text{ppb} = \frac{\text{W solute (g)}}{\text{W / V of solution (g / mL)}} \times 10^9$$

TRICK 1 ppm = 1 mg per litre of a dilute aqueous solution (density about 1 g/mL). 1 ppb = 1 microgram per litre. 1 % = 10^4 ppm.

Textbook: ppm and ppb (§1.4.7, §1.4.8)

Definitions

ppm: mass of solute present in one million (10^6) parts by mass of solution. Used when a very small amount of solute is in a large amount of solution.

$$\text{ppm} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6 = \frac{w}{w + w_0} \times 10^6$$

w = mass of solute, w_0 = mass of solvent. For very dilute solutions $w_0 \gg w$, so ppm ≈ $(w/w_0) \times 10^6$.
ppb: mass of solute present in one billion (10^9) parts by mass of solution: ppb = (mass of solute / mass of solution) × 10^9.

Solved examples

Textbook example: ppm of chloride TEXTBOOK §1.4.7 **Q.** 250 mL of a solution contains 10 mg of Cl^-. The density of the solution is 1.00 g/mL. Find the concentration in ppm.

Mass of solute = 10 mg = 10×10^{-3} g = 0.01 g.
Mass of solution = volume × density = 250 × 1.00 = 250 g.

$$\text{ppm} = \frac{0.01}{250} \times 10^6 = 40$$

Answer: 40 ppm

Textbook example: ppb of lead TEXTBOOK §1.4.8 **Q.** 500 mL of a solution contains 0.01 mg of Pb^{2+}. The density of the solution is 1.00 g/mL. Find the concentration in ppb.

Mass of solute = 0.01 mg = 0.01×10^{-3} g = 10^{-5} g.
Mass of solution = 500 × 1.00 = 500 g.

$$\text{ppb} = \frac{10^{-5}}{500} \times 10^9 = 20$$

Answer: 20 ppb

PART 8 Numericals on concentration terms

How to attack a concentration numerical

Solving habits

1. Write "Given" and "To find" first. Write WHAT the question asks.
2. Convert everything to the same unit before comparing concentrations (g/L, or M, or N). You cannot compare an apple with a person.
3. Cross-check by a second route; any correct formula gives the same answer. Do not do extra work: answer exactly what is

asked.
4. If only "%" is given and nothing else, take it as % w/v.

Textbook: conversion formulae ("Some Important Formulae")

Percentage and density forms

Derivation of the % w/w form. 1 L of solution of specific gravity d weighs 1000 d g. If it is x % w/w, solute = (x/100) × 1000 d = 10 x d g per litre. So g/L = 10 × % w/w × d, and dividing by E or M_w:

$$N = \frac{\%\ w/w \times d \times 10}{E} \qquad M = \frac{\%\ w/w \times d \times 10}{M_w}$$

$$N = \frac{\%\ w/v \times 10}{E} \qquad M = \frac{\%\ w/v \times 10}{M_w} \qquad \% = \frac{g/L}{10}$$

$$N = \frac{g/L}{E} \qquad M = \frac{g/L}{M_w} \qquad g/L = N \times E = M \times M_w$$

In CGS, density, specific gravity and relative density are numerically equal.

Solved examples (each with its textbook / manual twin)

Example 8.1 **Q.** Convert 2.5 M H_3PO_4 into normality.

$n(H_3PO_4) = 3$.

$$N = M \times n = 2.5 \times 3 = 7.5$$

Answer: 7.5 N

Example 8.2 NEB PYQ 2069 Set B **Q.** Find the g/L of 4.9 M H_2SO_4.

$$g/L = M \times M_w = 4.9 \times 98 = 480.2$$

Answer: 480.2 g/L (480.2 g H_2SO_4 in 1 L of solution)

Example 8.3 NEB PYQ 2066 Q17 NEB PYQ MCQ 11 **Q.** What is the normality of 20 cc of 2 M H_3PO_4?

$N = M \times n = 2 \times 3 = 6$ N. The 20 cc is irrelevant to normality.

Answer: 6 N

NEB PYQ MCQ 11 1 M H_3PO_4 = 3 N, option (c).

Example 8.4 NEB PYQ 2069 Set B Q2 **Q.** What mass of Na_2CO_3 is needed to prepare 50 cc of its semi-normal solution?

N = 0.5; E = 106/2 = 53; V = 50 mL (cc = mL = cm^3).

$$W = \frac{N \times E \times V(mL)}{1000} = \frac{0.5 \times 53 \times 50}{1000} = 1.325$$

Answer: 1.325 g

Example 8.5 TEXTBOOK §1.4 Solved Q2 **Q.** 5.3 g Na_2CO_3 is dissolved in 500 mL of solution. Find the molarity (and normality).

moles = 5.3/106 = 0.05; V = 0.5 L.

$$M = \frac{0.05}{0.5} = 0.1$$

$N = M \times n = 0.1 \times 2 = 0.2$. Direct route: $N = 5.3 \times 1000/(53 \times 500) = 0.2$ (agrees).

Answer: 0.1 M; 0.2 N

Example 8.6 TEXTBOOK §1.7 Solved Numerical 5 NEB PYQ 2072 Set D Q2 **Q.** Which has the higher concentration and why? (a) 80 g/L NaOH or 3 M NaOH. (b) 5.3 g/L Na_2CO_3 or N/10 Na_2CO_3.

(a) Convert 80 g/L to molarity: $M = (g/L)/M_w = 80/40 = 2$ M. Compare 2 M with 3 M: 3 M NaOH is higher.
(b) $E(Na_2CO_3) = 53$. $N = (g/L)/E = 5.3/53 = 0.1$ N = N/10. Both have the same concentration.
Reason to write: concentrations can be compared only in the same unit.

Answer: (a) 3 M is higher (b) equal

Example 8.7 NEB PYQ 2077 Set W Q9 **Q.** Calculate the molality of 1 L of 93 % (w/v) H_2SO_4 of density 1.84 g/mL.

93 % w/v: 93 g in 100 mL, so in 1000 mL there is 930 g H_2SO_4.
Mass of solution = density × volume = 1.84 × 1000 = 1840 g.
Mass of solvent = 1840 − 930 = 910 g = 0.910 kg.
Moles of H_2SO_4 = 930/98 = 9.490.

$$m = \frac{9.490}{0.910} = 10.43$$

Answer: 10.43 m

Marking note: the manual prints 10.41 by rounding 9.49 down to 9.48 first. Carry the digits: 10.43.

Example 8.8 NEB PYQ 2072 Supp Q25 **Q.** A solution of conc. HCl contains 38 % HCl by mass, density 1.19 g/cc. (i) Find its molarity. (ii) What volume of it neutralises 1 L of 0.1 M NaOH?

$$\text{(i) } M = \frac{\%\ w/w \times d \times 10}{M_w} = \frac{38 \times 1.19 \times 10}{36.5} = 12.39$$

(ii) HCl: n = 1 so N = 12.39. NaOH 1000 mL of 0.1 N.

$$V = \frac{1000 \times 0.1}{12.39} = 8.07 \text{ mL}$$

Answer: 12.39 M; 8.07 mL

Example 8.9 NEB PYQ 2071 Supp Q2 (same idea) **Q.** Calculate the molarity of a 5 % H_2SO_4 solution (only % is given, so take w/v).

5 g in 100 mL. moles = 5/98 = 0.0510; V = 0.1 L.

$$M = \frac{0.0510}{0.1} = 0.51$$

Answer: 0.51 M

Example 8.10 VERY IMPORTANT **Q.** A solution contains 1 mol alcohol and 4 mol water. Find the mole fraction of water and of alcohol.

Total = 1 + 4 = 5.

$$\chi_{water} = \frac{4}{5} = 0.8 \qquad \chi_{alcohol} = \frac{1}{5} = 0.2$$

Check: 0.8 + 0.2 = 1.

Answer: 0.8; 0.2

More PYQ numericals on concentration

NEB PYQ NEB-2083 Q2 **Normality of 5 % Na_2CO_3** **Q.** What is the normality of 5 % Na_2CO_3 solution? A) 0.94 N B) 0.98 N C) 0.99 N D) 1.01 N

5 % w/v = 50 g/L. E = 53.

$$N = \frac{50}{53} = 0.943$$

Answer: A) 0.94 N

SXC PYQ *2T-2024 Q13c* **Molality of liquor ammonia** **Q.** Liquor ammonia is 25 % (w/v) NH_3 with specific gravity 0.91. Find its molality. [3]

25 % w/v: 25 g NH_3 in 100 mL, so 250 g in 1 L. Mass of 1 L solution = 0.91 × 1000 = 910 g.
Solvent = 910 − 250 = 660 g = 0.660 kg. Moles NH_3 = 250/17 = 14.71.

$$m = \frac{14.71}{0.660} = 22.28$$

Answer: 22.28 m

NEB PYQ 2080 GIE Set B Q20 **Volume and normality of solution containing 23 g HNO_3** **Q.** Conc. HNO_3 is 69 % by mass, density 1.41 g/cm^3. Calculate the volume and normality of the solution containing 23 g HNO_3. [4]

69 g HNO_3 is in 100 g solution. So 23 g HNO_3 is in 100 × 23/69 = 33.33 g solution.

$$V = \frac{33.33}{1.41} = 23.64 \text{ cm}^3$$

$$N = \frac{69 \times 1.41 \times 10}{63} = 15.44$$

Answer: 23.64 cm^3; 15.44 N

NEB PYQ 2079 Set H Q21b **Molarity and molality of a dibasic acid solution** **Q.** A solution of a dibasic acid has 17.7 g per litre; density 1.0077 g/mL; M_w = 118. Calculate (i) molarity (ii) molality.

$$\text{(i) } M = \frac{17.7}{118} = 0.15$$

(ii) mass of 1 L solution = 1007.7 g; solvent = 1007.7 − 17.7 = 990 g = 0.990 kg.

$$m = \frac{0.15}{0.990} = 0.1515$$

Answer: 0.15 M; 0.15 m (0.1515)

NEB PYQ MCQ 13 **MCQ: normality of 3.65 g HCl per litre** **Q.** What is the normality of a solution containing 3.65 g HCl in 1 L? a) 1 N b) 0.5 N c) 0.1 N d) 2 N

$$N = \frac{3.65}{36.5 \times 1} = 0.1$$

Answer: c) 0.1 N

NEB PYQ MCQ 16 **MCQ: molar concentration of an antacid** **Q.** An antacid tablet containing 0.50 g $NaHCO_3$ is dissolved in 250 mL water. Molar concentration? a) 0.006 M b) 0.012 M c) 0.024 M d) 0.048 M

M_w = 84; moles = 0.50/84 = 0.00595.

$$M = \frac{0.00595}{0.250} = 0.0238$$

Answer: c) 0.024 M

PART 9 Principle of volumetric analysis; law of equivalence; normality equation; dilution; mixing; purity

Law of equivalence and the normality equation

Key points and tricks

Law of equivalence: if A and B react completely, g-eq of A = g-eq of B. True for any complete reaction, not only acid-base.
Normality equation: g-eq = N × V. For acid (1) and base (2): $N_1V_1 = N_2V_2$. Valid also for dilution (adding water does not

change g-eq of solute). V may stay in mL on both sides: do not waste time converting.

VERY IMPORTANT Exam: "Deduce $N_1V_1 = N_2V_2$". Write: "Let us consider reaction A + B → product. According to the law of chemical equivalence, substances react in proportion to their equivalent weights, so g-eq of acid = g-eq of base …"

Mixing (resultant normality N_R):

same type (all acids or all bases): $N_R = \dfrac{N_1V_1 + N_2V_2 + N_3V_3 + \dots}{V_1 + V_2 + V_3 + \dots}$

acid + base: $N_R = \dfrac{N_1V_1 - N_2V_2}{V_1 + V_2}$ (1 = acid: positive ⇒ ACIDIC; negative ⇒ BASIC)

% purity = (weight of pure substance / weight of impure sample) × 100. It is NOT % w/w. Only the pure part reacts; the sandy residue does not.

Textbook: law of equivalence and the principle (§1.6)

Law of equivalence

Substances react together in equal numbers of gram equivalents to produce the same number of gram equivalents of product. For A + B → C + D, at the equivalence point g-eq of A = g-eq of B = g-eq of C = g-eq of D.

1 g-eq acid = 1 g-eq base; 1000 cc of 1 N acid = 1000 cc of 1 N base; 100 cc of 10 N acid = 2000 cc of N/2 base; 50 cc of 20 N acid = 5000 cc of N/5 base.

$$V_A \times N_A = V_B \times N_B \quad \text{i.e.} \quad V_1N_1 = V_2N_2$$

The normality equation is the working formula of volumetric analysis.

Derivation of the normality equation (NEB 2081 Set P/Q Q20a; 2080 Set G/H Q20)

1. Take a neutralisation A + B → product (A = acid, B = base).
2. At the equivalence point: number of g-eq of A = number of g-eq of B ... (i)
3. From the definition N = g-eq / V(L), so g-eq = N × V(L) ... (ii)
4. Using (i) in (ii): $N_A \times V_A = N_B \times V_B$.
5. Since V is in the same unit on both sides (mL or L), $N_1V_1 = N_2V_2$. It holds for any acid-base neutralisation.

Why not always used for molarity (2080 G/H iii): normality depends on the n-factor, which changes with the reaction; molarity does not. The equation $M_1V_1 = M_2V_2$ is valid only when the acid and base have the same n-factor ratio (1:1).

Principle of volumetric analysis: four rules (§1.6)

1. Acid and base of equal normality neutralise each other in equal volumes. 100 mL 1 N HCl = 100 mL 1 N NaOH.
2. Volume is inversely proportional to normality. 50 mL N/2 HCl = 250 mL N/10 HCl; 100 mL N/10 HCl = 10 mL 1 N HCl.
3. Substances react in equivalent proportions: 0.5 g-eq HCl = 0.5 g-eq NaOH.
4. If acid and alkali neutralise, $N_1V_1 = N_2V_2$ holds.

Textbook formulae for mixing (item 17, "Some Important Formulae"): V_a mL of N_a acid + V_b mL of N_b base: neutral if $V_aN_a = V_bN_b$; acidic if $V_aN_a > V_bN_b$, $N = (V_aN_a - V_bN_b)/(V_a + V_b)$; basic if $V_bN_b > V_aN_a$, $N = (V_bN_b - V_aN_a)/(V_a + V_b)$.

Solved examples: dilution

Example 9.1 NEB PYQ 2077 Set V Q3 **VERY IMPORTANT** **Q.** How much water must be added to 50 mL of semi-normal NaOH to make it exactly decinormal?

$N_1 = 0.5$, $V_1 = 50$ mL; $N_2 = 0.1$, $V_2 = ?$

$N_1V_1 = N_2V_2 \Rightarrow 0.5 \times 50 = 0.1 \times V_2 \Rightarrow V_2 = 250$ mL (final volume).

Water added = $V_2 - V_1 = 250 - 50 = 200$ mL.

Answer: 200 mL

NEB PYQ MCQs 20 mL 2 N HCl to decinormal: $V_2 = 400$, add 380 mL (b). 10 mL 10 N HCl to decinormal: $V_2 = 1000$, add 990 mL (a); same as SXC 1T-2025 Q6. 100 mL 0.5 N to 0.2 N: add 150 mL (a). 400 mL N/10 HCl evaporated to 2 N: $V_2 = 20$, evaporate 380 mL (c).

Textbook Solved Numerical 3 (with normality factor) TEXTBOOK §1.7 Solved Num 3 NEB PYQ 2070 Set D Q2 **Q.** What volume of water must be added to 500 mL of 2 N (f = 0.98) Na_2CO_3 to make it exactly N/10?

Exact $N_1 = 2 \times 0.98 = 1.96$ N.

$500 \times 1.96 = V_2 \times 0.1 \Rightarrow V_2 = 9800$ mL.

Water added = 9800 − 500 = 9300 mL.

Answer: 9300 mL

Concentrated HNO_3 needed TEXTBOOK §1.7 Solved Num 7 NEB PYQ 2075 Set B Q25 NEB PYQ 2072 Set D Q25 **VERY IMPORTANT** **Q.** How many mL of conc. HNO_3 (specific gravity 1.41, 69 % by mass) are needed to prepare 500 mL of 0.5 N HNO_3?

Step 1: normality of the conc. acid. $E(HNO_3) = 63$.

$$N_1 = \frac{69 \times 1.41 \times 10}{63} = 15.44$$

Step 2: dilution $N_1V_1 = N_2V_2$: $15.44 \times V_1 = 0.5 \times 500$

$$V_1 = \frac{250}{15.443} = 16.19 \text{ mL}$$

Answer: 16.19 mL

Dilute conc. H_2SO_4 and find mass of water *SXC PYQ 2T-2024 Q12* NEB PYQ 2071 Supp Q25 **Q.** What is the normality factor? What volume of 95 % H_2SO_4 (d = 1.85 g/cc) and what mass of water make 100 cc of 15 % H_2SO_4 (d = 1.1 g/cc)? [1+4]

f = weight of substance taken / weight to be taken.

NaOH added to H_2SO_4; which litmus? NEB PYQ 2080 Set G/H Q20iv NEB PYQ 2068 Q30 **Q.** 4 g NaOH is added to 20 cc of 2 N H_2SO_4 and diluted to 1 L. (a) Which litmus paper would you use and why? (b) Normality of the dilute solution.

Acid: 20 × 2 = 40 meq. Base: 4/40 = 0.1 g-eq = 100 meq. Excess base 60 meq, so BASIC.

(a) Use RED litmus: it turns blue in a basic solution.

$$\text{(b) } N = \frac{60}{1000} = 0.06 \qquad (\text{g/L of NaOH} = 0.06 \times 40 = 2.4)$$

Answer: Basic; red litmus turns blue; 0.06 N

Mixing NaOH and H_2SO_4 given as molarity NEB PYQ 2066 Q25 NEB PYQ 2080 Set J Q12 **Q.** Find the resulting normality of 20 mL of 0.8 M NaOH mixed with 25 mL of 0.4 M H_2SO_4.

NaOH: N = 0.8; meq = 20 × 0.8 = 16. H_2SO_4: N = 0.8; meq = 25 × 0.8 = 20.

$$N_R = \frac{20 - 16}{20 + 25} = \frac{4}{45} = 0.0889 \text{ (acidic)}$$

Answer: 0.089 N (acidic)

Small excess of acid NEB PYQ 2072 Set E Q2 **Q.** 0.53 g Na_2CO_3 is added to 101 mL of 0.1 N H_2SO_4. Acidic, basic or neutral?

Na_2CO_3: 0.53/53 = 0.01 g-eq = 10 meq. Acid: 101 × 0.1 = 10.1 meq. Acid is in slight excess (0.1 meq).

Answer: Acidic

MCQs on mixing NEB PYQ MCQs 15, 17, 19 **Q.** (a) 100 mL 0.5 M H_2SO_4 + 0.1 L 1 M HCl: N? (b) 10 mL 1 M H_2SO_4 + 25 mL 0.4 N HCl: N? (c) 20 mL 1 N HCl + 30 mL 0.5 N NaOH: N?

(a) 0.5 M H_2SO_4 = 1 N. (100 + 100)/200 = 1 N.

(b) 1 M H_2SO_4 = 2 N. (20 + 10)/35 = 0.857 N.

(c) (20 − 15)/50 = 0.1 N, acid left, so 0.1 N HCl.

Answer: (a) 1 N (b) 0.857 N (c) 0.1 N HCl

Antacid problem TEXTBOOK §1.7 Solved Num 19 *SXC PYQ PB-2025 Q20c* VERY IMPORTANT **Q.** Three antacid tablets reduce stomach acid from N/10 to N/100 in 1000 cc fluid. Find the base per tablet (E of base = 100).

Acid before: 1000 × 0.1 = 100 meq. After: 1000 × 0.01 = 10 meq. Neutralised = 90 meq = 0.09 g-eq.

Mass of base = 0.09 × 100 = 9 g for three tablets.

$$\text{Per tablet} = \frac{9}{3} = 3 \text{ g}$$

Answer: 3 g

Solved examples: % purity and mixtures

Example 9.5 TEXTBOOK §1.7 Solved Num 4 **Q.** x g of Na_2CO_3 reacts completely with 50 mL of N/2 HCl. Find x.

g-eq Na_2CO_3 = g-eq HCl. Use W/E for the weight and NV/1000 for the volume:

$$\frac{x}{53} = \frac{0.5 \times 50}{1000} \Rightarrow x = 1.325$$

Answer: 1.325 g

TEXTBOOK Solved Num 4 x g Na_2CO_3 with 20 mL of 1 N HCl: x/53 = 20/1000, x = 1.06 g.

Example 9.6 VERY IMPORTANT **Q.** 1 g of an ordinary limestone ($CaCO_3$) is dissolved in 16.6 cc of 0.92 N HCl, leaving a sandy residue. Find the % purity.

Only pure $CaCO_3$ reacts: g-eq pure $CaCO_3$ = g-eq HCl = 0.92 × 16.6/1000 = 0.01527.

E($CaCO_3$) = 100/2 = 50. W(pure) = 0.01527 × 50 = 0.7636 g.

$$\%\text{ purity} = \frac{0.7636}{1} \times 100 = 76.4$$

Answer: 76.4 %

Mass of 90 % pure $CaCO_3$ TEXTBOOK §1.7 Solved Num 6 NEB PYQ 2071 Set C Q2 **Q.** What mass of 90 % pure $CaCO_3$ neutralises 2 L of decinormal HCl?

2 L of 0.1 N = 200 meq = 0.2 g-eq. Pure $CaCO_3$ = 0.2 × 50 = 10 g.

$$\text{Impure mass} = \frac{10 \times 100}{90} = 11.11$$

Answer: 11.11 g

% purity of commercial H_2SO_4 TEXTBOOK §1.7 Solved Num 8 NEB PYQ 2072 Set E Q26 **Q.** A commercial H_2SO_4 has specific gravity 1.8. 10 mL is diluted to 1 L; 10 mL of the dilute acid needs 30 mL of N/10 NaOH. Find the % purity.

Dilute acid: N = 30 × 0.1/10 = 0.3. Commercial: 10 × N = 1000 × 0.3 ⇒ N = 30.

$$30 = \frac{\% \times 10 \times 1.8}{49} \Rightarrow \% = \frac{30 \times 49}{18} = 81.67$$

Answer: 81.67 %

Mixture of NaCl and Na_2CO_3 TEXTBOOK §1.7 Solved Num 16 **Q.** 1.0 g of a mixture of NaCl and Na_2CO_3 in 250 mL; all of it needs 50 mL N/10 HCl. Find the composition.

Only Na_2CO_3 reacts: 50 × 0.1 = 5 meq. W = 5 × 53/1000 = 0.265 g.

Na_2CO_3 = 26.5 %; NaCl = 100 − 26.5 = 73.5 %.

Answer: 26.5 % Na_2CO_3; 73.5 % NaCl

Mixture of $CaCO_3$ and $MgCO_3$ TEXTBOOK §1.7 Solved Num 17 **Q.** 1.97 g of a $CaCO_3$ + $MgCO_3$ mixture is dissolved in 400 mL of 0.15 N HCl and made to 300 mL; 15 mL of it needs 18 mL of 0.05 N NaOH. Find the composition.

HCl taken = 400 × 0.15 = 60 meq. 15 mL needs 18 × 0.05 = 0.9 meq, so 300 mL needs 0.9 × 20 = 18 meq excess. Used by mixture = 60 − 18 = 42 meq.
E($CaCO_3$) = 50, E($MgCO_3$) = 84/2 = 42. Let $CaCO_3$ = x g: x/50 + (1.97 − x)/42 = 0.042.
(0.02 − 0.02381)x = 0.042 − 0.04690 ⇒ x = 1.2875 g $CaCO_3$; $MgCO_3$ = 0.6825 g.
$CaCO_3$ = 1.2875/1.97 × 100 = 65.35 %; $MgCO_3$ = 34.65 %.
Answer: 65.35 % $CaCO_3$; 34.65 % $MgCO_3$

Marking note: the textbook prints 64.97 % / 35.03 % (it rounds 23.8 and x = 1.28 early).

NH_4Cl in a mixture with NaCl TEXTBOOK §1.7 Solved Num 18 **Q.** 0.5 g NH_4Cl + NaCl is boiled with 25 mL of 0.95 N NaOH until NH_3 is expelled; the residue is made to 100 mL; 10 mL of it neutralises 17 mL of 0.1 N (f = 0.998) HCl. Find % NH_4Cl.

NaOH taken = 25 × 0.95 = 23.75 meq. 10 mL needs 17 × 0.1 × 0.998 = 1.6966 meq, so 100 mL has 16.966 meq left.
Used by NH_4Cl = 23.75 − 16.966 = 6.784 meq. E(NH_4Cl) = 53.5: mass = 6.784 × 53.5/1000 = 0.3629 g.

$$\% \ NH_4Cl = \frac{0.3629}{0.5} \times 100 = 72.58$$

Answer: 72.58 %

Equivalent and molecular mass of a dibasic acid TEXTBOOK §1.7 Solved Num 15 NEB PYQ 2063 Q18 **Q.** 7.35 g of a dibasic acid is made to 250 mL; 25 mL needs 15 mL of 1 N NaOH. Find E and M_w. (NEB: 0.315 g of a dibasic acid needs 50 mL of decinormal NaOH. Find M_w.)

25 × N = 15 × 1 ⇒ N = 0.6. E = 7.35 × 1000/(0.6 × 250) = 49. M_w = 49 × 2 = 98.
NEB 2063: meq NaOH = 50 × 0.1 = 5; E = 0.315 × 1000/5 = 63; M_w = 63 × 2 = 126.
Answer: E = 49, M = 98; M = 126

Equimolar mixture of Na_2CO_3 and $NaHCO_3$ NEB PYQ 2076 GIE Set A Q23 **Q.** How many mL of 0.1 M HCl completely react with 1 g of a mixture of Na_2CO_3 and $NaHCO_3$ containing equimolar amounts?

Let n mol of each: 106n + 84n = 1 ⇒ n = 1/190 = 0.005263 mol.
HCl needed: Na_2CO_3 + 2HCl and $NaHCO_3$ + HCl, so moles HCl = 2n + n = 3n = 0.015789.

$$V = \frac{0.015789}{0.1} = 0.1579 \text{ L} = 157.9 \text{ mL}$$

Answer: 157.9 mL

Marking note: the manual's printed answer is 202.6 mL, but its own two terms add to 0.01579 g-eq, which gives 157.9 mL.

PART 10 Back titration; metal + acid problems

Back titration ("very, very, very important; asked all the time")

VERY IMPORTANT **Back titration and metal + acid problems**

Normal case: A + B react completely, g-eq A = g-eq B. **Back titration:** A is reacted with an EXCESS of B. A is used up, some B is left. The left-over B is neutralised by a second substance X, so B is used up by TWO things:

g-eq of B (taken) = g-eq of A + g-eq of X

Typical case (metal + acid): metal is added to excess acid; the left-over acid is neutralised by a base (usually NaOH):

g-eq acid (initial) = g-eq metal + g-eq base

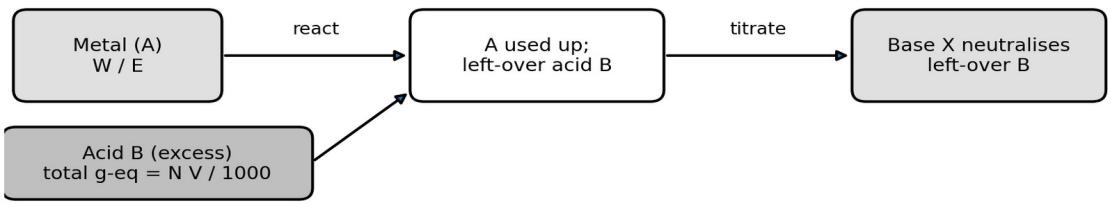
Use **W/E** for the metal; **N × V(mL)/1000** for acid and base.

TRICK If the solution is DILUTED and only a part (aliquot) is titrated, scale the base volume up to the whole solution: 25 mL of a 100 mL solution used V mL, so the whole needs V × (100/25) = 4V. Concentration does not change when you take a portion. Never skip this step.

E of a metal = atomic weight / valency. "Divalent" means valency 2 (trivalent 3, tetravalent 4). Atomic mass = E × valency.

Write what is given step by step, keeping the metal part and the base part separate.

g-eq of acid taken = g-eq of metal + g-eq of base used for the excess



Back titration scheme (redrawn for these notes)

Solved examples (all in the form: g-eq acid = g-eq metal + g-eq base)

Example 10.1 VERY IMPORTANT NEB PYQ 2078 Set D Q23 **Q.** 0.4 g of a divalent metal is dissolved in 50 cc of 0.64 N HCl and diluted to 100 cc. 25 cc of this solution needs 27.3 cc of 0.11 N NaOH for complete neutralisation. Find the atomic mass of the metal.

Whole 100 cc needs 27.3 × (100/25) = 109.2 cc of 0.11 N NaOH (concentration unchanged, so scale the volume).
g-eq HCl = 0.64 × 50/1000 = 0.032. g-eq NaOH = 0.11 × 109.2/1000 = 0.012.

PART 11 Water of crystallisation; Zn / H_2 gas purity problem

Water of crystallisation ($Na_2CO_3.xH_2O$)

Key points and tricks

Complete neutralisation: g-eq of the hydrated salt = g-eq of HCl. Left side W/E; right side NV/1000.
E of $Na_2CO_3.xH_2O$ = M_w/n with $M_w = 106 + 18x$ and **n = 2** (n comes from Na_2CO_3 only; the water does NOT change n).

$$E = \frac{106 + 18x}{2} = 53 + 9x$$

TRICK Find E experimentally (E = W ÷ g-eq), then solve 53 + 9x = E. Answer to the nearest whole number (x is an integer: 1, 2, 5, 7, 10).

Solved examples

Example 11.1 NEB PYQ 2064 Q30iii **VERY IMPORTANT** **Q.** 0.715 g of $Na_2CO_3.xH_2O$ needs 20 mL of semi-normal HCl for complete neutralisation. Find x.

g-eq HCl = 0.5 × 20/1000 = 0.01. So 0.715/E = 0.01 ⇒ E = 71.5.
53 + 9x = 71.5 ⇒ 9x = 18.5 ⇒ x = 2.06 ≈ 2.
The salt is $Na_2CO_3.2H_2O$.
Answer: x = 2

0.70 g in 100 mL, titrate 20 mL NEB PYQ 2079 GIE Set B Q13 NEB PYQ 2081 Set P/Q Q20c **VERY IMPORTANT** **Q.** 0.70 g of $Na_2CO_3.xH_2O$ is dissolved and made to 100 mL. 20 mL needs 19.8 mL of N/10 HCl. Name the indicator and find x.

Indicator: methyl orange (weak base Na_2CO_3 + strong acid HCl).
Dilute: 100 mL needs 19.8 × 100/20 = 99 mL of 0.1 N HCl = 9.9 meq = 0.0099 g-eq.

$$E = \frac{0.70}{0.0099} = 70.71 \quad \Rightarrow \quad 53 + 9x = 70.71 \quad \Rightarrow \quad x = 1.97 \approx 2$$

Answer: x = 2 ($Na_2CO_3.2H_2O$)

Soda crystals: find x and % anhydrous Na_2CO_3 *SXC PYQ 1T-2024 Q12b* **VERY IMPORTANT** **Q.** 3.896 g of soda crystal ($Na_2CO_3.xH_2O$) is dissolved in 250 mL. 20 mL needs 21.8 mL of N/10 H_2SO_4. Find the % of anhydrous Na_2CO_3 and x. [3]

$$N_{soln} = \frac{0.1 \times 21.8}{20} = 0.109$$

g-eq in 250 mL = 0.109 × 0.250 = 0.02725. Anhydrous Na_2CO_3 = 0.02725 × 53 = 1.4443 g.

$$\% Na_2CO_3 = \frac{1.4443}{3.896} \times 100 = 37.07$$

E of crystal = 3.896/0.02725 = 142.97 ≈ 143, M_w = 286 = 106 + 18x ⇒ x = 10 (washing soda).
Answer: 37.07 %; x = 10

The Zn / H_2 gas purity problem

Key points and tricks

Reaction: $Zn + H_2SO_4(dil.) \rightarrow ZnSO_4 + H_2$. 1 mol Zn gives 1 mol H_2. H_2SO_4 is in excess, so zinc is the limiting reagent.
Convert the gas volume to moles with PV = nRT:

$$n = \frac{PV}{RT}$$

P in atm (760 mmHg = 1 atm), V in litres, T in kelvin, R = 0.0821 L atm mol^{-1} K^{-1} (8.314 with joules). "Last question of the type, asked recently."

Example 11.2 **VERY IMPORTANT** NEB PYQ 2075 GIE Q23 **Q.** 12 g of commercial zinc reacts with excess dilute H_2SO_4. The H_2 collected is 4.2 L at 570 mmHg and 279 K. Find the % purity of zinc.

P = 570/760 = 0.75 atm; V = 4.2 L; T = 279 K.

$$n(H_2) = \frac{0.75 \times 4.2}{0.0821 \times 279} = \frac{3.15}{22.906} = 0.1375 \text{ mol}$$

Moles Zn = moles H_2 = 0.1375. Mass of pure Zn = 0.1375 × 65 = 8.94 g (Zn = 65).

$$\% \text{ purity} = \frac{8.94}{12} \times 100 = 74.49$$

Answer: 74.49 % (about 74.5 %)

PART 12 Titration process and terminology

Titration procedure

Key points and tricks

Titration = finding the concentration of an UNKNOWN solution by reacting it with a solution of KNOWN concentration. Generally acid in the burette and base in the flask (either way is allowed).

1. Pipette 10 mL of NaOH (unknown) into a conical flask.
2. Add 2-3 drops of phenolphthalein: solution turns PINK (base).
3. Fill the burette with HCl of known concentration (say N/10).
4. Open the stopcock slowly; add HCl drop by drop, shaking the flask.
5. When the pink colour disappears (colourless), STOP: that is the end point. Read the burette. Repeat to get three **concurrent readings** (readings that agree) and take the matching value as the main reading.

6. Use $N_1V_1 = N_2V_2$ to find the unknown normality.

TRICK **T for Top, T for known**: TITRANT ends with t, it is the KNOWN solution in the burette. **D for Down, D for determined**: TITRAND ends with d, it is the solution whose concentration is DETERMINED, in the conical flask. Which indicator suits which pair is explained below (pH curves); the full theory belongs to Ionic Equilibrium.

Textbook: definitions of titration terms (§1.7)

Titration and its terms

Titration: the process of determining the concentration of an unknown solution by finding the volume of it required to react completely with a known volume of a standard solution.

Indicator (§1.7.3): a chemical substance used in titration to indicate the completion of the reaction by a distinct colour change. It is a weak organic acid or base that changes colour because its structure changes. **Phenolphthalein** is a weak organic acid: colourless in acid, pink in alkali (the molecule is colourless, the ion is pink). **Methyl orange** is a weak base: red in acid, yellow in alkali.

End point (§1.7.4): the point observed in the titration at which the reaction is completed, shown by a sharp colour change of the indicator. It is the experimental point.

Equivalence point (§1.7.5): the point at which the added titrant is chemically equivalent to the titrand (the two solutions are used up in stoichiometric proportions). It is not indicated by the indicator. It generally comes just before the end point.

Unknown solution (§1.7.6): the solution whose concentration is to be determined. **Standard solution (§1.7.7):** a solution of known concentration used to determine the concentration of an unknown solution.

Titration error (§1.7.8): the difference between the equivalence point and the measured end point. To minimise it, choose the indicator so that the end point coincides with the equivalence point.

Titrant (§1.7.9): the solution used to determine the unknown, usually the standard solution delivered from the burette. **Titrand (§1.7.10):** the solution of unknown concentration, usually taken in the flask with a pipette.

Neutral point (§1.7.11): the point in an acid-base titration where H^+ and OH^- are approximately balanced.

	End point	Equivalence point
1	The point at which the reaction is complete as shown by a sharp colour change of the indicator	The point at which the two solutions are used up in stoichiometric proportions
2	Comes AFTER the equivalence point	Comes BEFORE the end point
3	Observed in the burette-pipette system	Not observed (cannot be seen by the naked eye)
4	Titrant is not exactly equivalent to titrand (moles of titrant slightly exceed analyte)	Titrant is chemically equivalent to titrand (moles of titrant = moles of analyte)
5	Weak acids can have only one end point	Weak acids can have multiple equivalence points
6	Must be in an acid-base solution (NEB 2075 GIE)	May or may not be in an acid-base solution

Textbook §1.7.5 table (rows 1-5) with the NEB 2075 GIE / 2081 Set R row 6. Tags: **VERY IMPORTANT** *NEB 2080 GIE B, 2081 R, 2075 GIE, 2069 A, 2064.*

Titrant	Titrand
Taken in the burette	Substance being analysed (flask)
Mostly its concentration is exactly known	Mostly its concentration is unknown

NEB PYQ 2076 GIE Set A Q2 *Distinguish between titrant and titrand [2].*

Textbook: experimental procedure and Fig. 1.1 (§1.7.13)

1. Wash the burette with detergent solution, then rinse with distilled water.
2. Take about 50 mL of the standard acid in a clean dry 250 mL beaker and the same amount of the base of unknown concentration in another.
3. Pour about 5 mL of the base into the burette and rinse it. Fill the burette to the zero mark.
4. Wash the pipette with water, then with the acid to be pipetted.
5. Suck the acid up to the mark and run it into a clean titration flask.
6. Add one or two drops of indicator.
7. Swirl the flask gently and add the base drop by drop until the indicator changes colour (end point).
8. Note the exact final burette volume.
9. Repeat the titration to get concurrent readings.

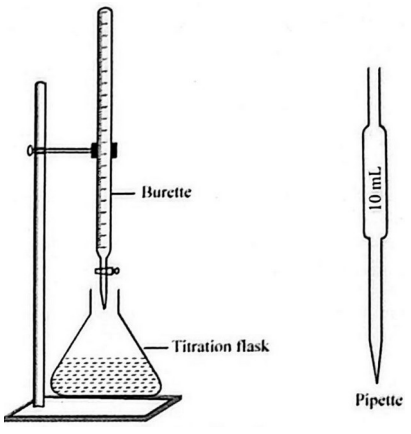


Fig. 1.1 Titration (Textbook, p. 19)

Applications (§1.7.14): quality control of soap, pharmaceuticals, other chemical products and biodiesel; impurities in manufactured chemicals; NEB 2080 GIE A: aspirin or vitamin C in tablets and purity of additives; 2079 Set H: salt content of food.

Indicator choice and the pH curve (Ionic Equilibrium link, included for the exams)

How an indicator is selected (NEB-2082 OR / PB-2025 Q20 OR [3]; SXC 1T-2024 Q2)

Near the equivalence point the pH changes suddenly over a few drops. An indicator is suitable only if its pH range lies inside this steep part of the curve; then end point ≈ equivalence point and the titration error is small.

Titration	pH at equivalence	Indicator	Examples
Strong acid + strong base	7	Either methyl orange or phenolphthalein	HCl + NaOH
Weak acid + strong base	> 7 (basic salt)	Phenolphthalein (8.3 - 10)	$(COOH)_2$ + NaOH, CH_3COOH + NaOH

quantity of reactant needed to react completely with another substance. NEB 2073 Supp: "importance of calculating f": to minimise experimental error.

Titration error	Normality factor
Difference between end point and equivalence point	Ratio of weight of solute taken to weight to be taken
Minimum when end point nearly coincides with equivalence point	Closest to 1 when weight taken ≈ weight to be taken
Not denoted by a symbol	Denoted by f

Solved examples

Example 13.1 (oxidation numbers) **Q.** Find the oxidation number of C in oxalic acid and of Mn in $KMnO_4$; hence the n-factor of each in the titration.

C in $H_2C_2O_4$: +3, going to +4 in CO_2: each C changes by 1, two C atoms: n = 2, E = 90/2 = 45 (anhydrous) or 126/2 = 63 (crystal).
Mn in $KMnO_4$: +7, going to +2: n = 5, E = 158/5 = 31.6.
Answer: C +3 (n = 2); Mn +7 (n = 5)

Normality factor of an oxalic acid solution ***SXC PYQ*** *1T-2024 Q21f* **VERY IMPORTANT** **Q.** A student weighs 0.65 g of oxalic acid (crystal) to make 100 mL of decinormal solution. Express the actual concentration using the factor. [2]

Weight to be taken = N × E × V/1000 = 0.1 × 63 × 100/1000 = 0.63 g.

$$f = \frac{0.65}{0.63} = 1.032$$

Actual normality = 0.1 × 1.032 = 0.1032 N, written N/10 (f = 1.032).
Answer: N/10 solution with f = 1.032 (actual 0.103 N)

Example 13.2 (using f in a titration calculation) **NEB PYQ** 2080 GIE A Q20 **TEXTBOOK** §1.7 Solved Num 3 **Q.** How is f used? Take 20 mL of N/10 HCl with f = 0.98 against NaOH.

Exact normality = 0.1 × 0.98 = 0.098 N. Use 0.098 (not 0.1) in $N_1V_1 = N_2V_2$.
Worked uses: Part 9 (lab-boy problem; 500 mL of 2 N, f = 0.98 → add 9300 mL) and Part 10 (acid with f = 1.01, base with f = 0.8).
Answer: N = proposed N × f

% $FeSO_4.7H_2O$ in a ferrous sulphate sample ***SXC PYQ*** *1T-2025 Q15* **VERY IMPORTANT** **Q.** Define decinormal solution. 25 g of ferrous sulphate sample is dissolved in water containing 20 mL of 2 N H_2SO_4 and made to 1 L. 25 mL of this solution needs 20 mL of decinormal $KMnO_4$. Find the % of $FeSO_4.7H_2O$. [1+4]

Decinormal solution: one-tenth g-equivalent of solute in 1 L of solution (N/10 = 0.1 N).
$Fe^{2+} \rightarrow Fe^{3+}$ (n = 1): E($FeSO_4.7H_2O$) = 278 (56 + 32 + 64 + 126 = 278).

$$25 \times N_{Fe} = 20 \times 0.1 \Rightarrow N_{Fe} = \frac{2}{25} = 0.08$$

g/L = N × E = 0.08 × 278 = 22.24 g in 1 L, i.e. in the whole 25 g sample.

$$\% = \frac{22.24}{25} \times 100 = 88.96$$

Answer: 88.96 %

PYQ drill: definitions in this part

NEB PYQ 2074 Supp Q2 **NEB PYQ** 2070 Supp Q2 **NEB PYQ** 2072 Supp Q25 **NEB PYQ** 2069 A **NEB PYQ** 2068 **Define acidimetry / alkalimetry** **Q.** Define acidimetry. Define alkalimetry. [1 each]

Acidimetry: determination of the concentration of an alkali by titrating it against a standard acid solution in the presence of an indicator.
Alkalimetry: determination of the concentration of an acid by titrating it against a standard alkali solution in the presence of an indicator.

NEB PYQ 2067 Q17 **NEB PYQ** 2072 Set C **NEB PYQ** 2073 Set D **NEB PYQ** 2068 **Define redox titration; give an example** **Q.** What is meant by redox titration? Write an example. Why is it so called? [1+1]

A titration involving oxidation and reduction together. Example: $2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 10CO_2 + 8H_2O$. $KMnO_4$ is the oxidising agent (reduced to $MnSO_4$) and oxalic acid the reducing agent (oxidised to CO_2), hence "redox".

NEB PYQ 2073 Supp Q2 **NEB PYQ** 2074 A/2073 C Q2 **NEB PYQ** 2076 Set C Q2 ***SXC PYQ*** *1T-2024 Q21* ***SXC PYQ*** *2T-2025 Q12a* **NEB PYQ** 2080 GIE A Q20 **Importance of normality factor; distinguish from titration error** **Q.** What is the normality factor? What is its importance in titration? Distinguish normality factor from titration error. [1+1+2]

f = weight of solute taken / weight to be taken. Importance: it corrects the strength of the solution for the weighing error and so minimises experimental error. For the difference see the table above (a ratio of weights vs a difference between two points of a titration).

NEB PYQ 2081 Set P/Q Q20b **NEB PYQ** 2079 Set H Q21a ***SXC PYQ*** *PB-2025 Q20a(ii)* **NEB PYQ** NEB-2082 Q20d **VERY IMPORTANT** **Why dil. H_2SO_4 and warming before titrating with $KMnO_4$** **Q.** Why is oxalic acid solution warmed by adding dilute H_2SO_4 before titrating with $KMnO_4$? [1]

Dilute H_2SO_4 supplies the H^+ (acidic medium) needed to reduce MnO_4^- to Mn^{2+}, since oxalic acid is too weak an acid to supply it. The solution is warmed to about 60 °C because the reaction between oxalic acid and $KMnO_4$ is slow at room temperature; the higher kinetic energy of the molecules increases the rate.

NEB PYQ 2079 Set D Q21a **NEB PYQ** 2079 Set H Q21a **Name the indicator in $KMnO_4$ vs oxalic acid; why redox? Q.** Name the indicator used in the titration of $KMnO_4$ against oxalic acid. Why is it called a redox titration?

No external indicator: $KMnO_4$ is a self-indicator (its own pink colour). $KMnO_4$ is the oxidising agent and oxalic acid the reducing agent, so it is redox.

PART 14 Primary and secondary standard substances and solutions

Primary and secondary standards

Key points and tricks

Primary standard substance: a substance from which a standard solution of exactly known concentration can be prepared directly by weighing. The solution made from it is a **primary standard solution**. Write several characteristics (more points = better marks):

1. Pure and dry (exactly the formula weight; free of impurities).
2. High molecular weight and hence high equivalent weight: reduces weighing error (weighing 1 kg is easier than 1 g; larger amounts give a smaller percent error).
3. Non-hygroscopic (does not absorb moisture from air).
4. Non-deliquescent (does not absorb so much water that it dissolves).
5. Non-efflorescent (does not lose its water of crystallisation in air).
6. Soluble in water.

Secondary standard substance: does not fully meet these conditions; its solution must be standardised by titration against a PRIMARY standard solution. To prepare a secondary standard solution: make a solution of the substance, then standardise it by titration against a primary solution.

Hygroscopic = absorbs water from air. **Deliquescent** = absorbs so much water that it dissolves and becomes liquid.

Examples (follow the textbook in the exam): primary: oxalic acid crystals, anhydrous Na_2CO_3, Mohr's salt, $AgNO_3$, KCl, $K_2Cr_2O_7$. Secondary: NaOH, KOH, HCl, H_2SO_4, HNO_3, $KMnO_4$. (Mohr's salt is written as a primary standard in the textbook and the NEB manual.)

Textbook: standard substances and solutions (§1.5)

Definitions and requisites (§1.5.1, §1.5.2)

Primary standard substance: pure, non-toxic, non-hygroscopic, non-deliquescent, stable in solid state or in solution for a sufficiently long time, and of high equivalent mass. Examples: oxalic acid crystals, sodium carbonate, Mohr's salt.

Requisites: (i) available in a pure state; (ii) non-toxic; (iii) not hygroscopic or deliquescent; (iv) high equivalent mass to minimise weighing error; (v) composition unchanged in solid state or in solution for a long time.

Primary standard solution: a standard solution prepared directly by dissolving a calculated mass of a primary standard substance in a definite volume of solution. Examples: N/10 Na_2CO_3, N/10 oxalic acid.

Secondary standard substance: cannot be used to prepare a standard solution directly by weighing. Examples: NaOH, HCl, HNO_3, $KMnO_4$. **Secondary standard solution:** a solution first standardised with a primary standard solution, then used to find an unknown. Examples: N/10 NaOH, N/10 KOH, N/10 $KMnO_4$, N/2 HNO_3.

Primary standard solution	Secondary standard solution
Prepared directly by dissolving a calculated mass of a primary standard substance in a definite volume	Prepared by dissolving a secondary standard substance in water and standardised with a primary standard solution
Strength does not change for a sufficiently long time	Strength may change slightly after a long time
Na_2CO_3, oxalic acid crystals, Mohr's salt (also $AgNO_3$, KCl in NEB answers)	$KMnO_4$, NaOH, HCl, HNO_3, H_2SO_4

TEXTBOOK §1.7 table **NEB PYQ** 2075 Set B Q2, 2071 Set C Q25, 2080 Set J Q12, NEB-2083 Q12a **VERY IMPORTANT**

Preparation of a primary standard solution (§1.5.3)

Example: 500 mL of N/10 oxalic acid from crystals. E = 126/2 = 63.

$$W = \frac{N \times E \times V(mL)}{1000} = \frac{0.1 \times 63 \times 500}{1000} = 3.15$$

Dissolve 3.15 g of oxalic acid crystals in water and make up to the mark of a 500 mL volumetric flask. (NEB-2083 Q12b uses exactly this data: 3.15 g in 500 mL gives 0.1 N.)

Solved examples

Example 14.1 (deduce $N_1V_1 = N_2V_2$; $KMnO_4$ vs oxalic acid) **NEB PYQ** NEB-2082 Q20 **VERY IMPORTANT** **Q.** Deduce the normality equation. N/10 oxalic acid (10 mL each time) is titrated with $KMnO_4$ from the burette: Expt 1: 0.0 to 11.5 mL; Expt 2: 0.0 to 11.0 mL; Expt 3: 0.0 to 11.0 mL. (a) Name the titration. (b) Equivalent weight of $KMnO_4$. (c) Normality of $KMnO_4$. (d) Why add dil. H_2SO_4 to the flask? (e) Titrant and titrand. [2+1+1+2+1+1]

Derivation: see Part 9 (g-eq acid = g-eq base, $N_1V_1 = N_2V_2$).

(a) Redox titration ($KMnO_4$ oxidant, oxalic acid reductant).

(b) $$E = \frac{158}{5} = 31.6$$ (acidic medium, Mn: +7 → +2)

(c) Concurrent (matching) readings are 11.0 and 11.0; take 11.0 mL (11.5 is rejected). N(oxalic) × V = N($KMnO_4$) × V:

$$N_{KMnO_\{4\}} = \frac{0.1 \times 10}{11.0} = 0.0909$$

(d) Dilute H_2SO_4 gives the acidic medium (H^+) needed for n = 5; oxalic acid is too weak an acid to supply it.

(e) Titrant = $KMnO_4$ (known strength in burette); titrand = oxalic acid (flask).

Answer: (a) redox (b) 31.6 (c) 0.0909 N (d) acid medium (e) titrant $KMnO_4$, titrand oxalic acid

Mass of $KMnO_4$ per litre NEB PYQ NEB-2083 Q12b VERY IMPORTANT **Q.** 3.15 g of oxalic acid crystal ($C_2H_2O_4.2H_2O$) is dissolved to make 500 mL. (i) Normality? (ii) The acid solution became permanently pink when 400 mL of $KMnO_4$ solution (E = 31.5) was added. What mass of $KMnO_4$ is in 1 L of the $KMnO_4$ solution? [1+2]

(i) $$N = \frac{3.15 \times 1000}{63 \times 500} = 0.1$$

(ii) Oxalic acid: $0.1 \times 500 = 50$ meq. At the end point 400 mL $KMnO_4$ holds 50 meq.

$$N_{KMnO_{4}} = \frac{50}{400} = 0.125$$

g/L = N × E = 0.125 × 31.5 = 3.94 g (E = 31.5 as given in the question).

Answer: (i) 0.1 N (ii) 3.94 g per litre

Model answer: PB-2025 Q20 (all parts) *SXC PYQ* *PB-2025 Q20* VERY IMPORTANT **Q.** (a) Oxalic acid crystal as primary standard: (i) define primary standard solution; (ii) why warmed with dil. H_2SO_4 before $KMnO_4$; (iii) E of $KMnO_4$ in acidic medium. (b) Volumes of N/2 and N/10 HCl to give 2 L of N/5 HCl. (c) Antacid: 3 tablets reduce acid from N/10 to N/100 in 1000 mL; base per tablet (E = 100). [3+2+3]

(a)(i) A solution of exactly known concentration made directly by dissolving a weighed amount of a primary standard substance in a definite volume. (ii) Medium and rate: see Part 13. (iii) 158/5 = 31.6.

(b) x L of N/2: $0.5x + 0.1(2 - x) = 0.4 \Rightarrow x = 0.5$ L of N/2 and 1.5 L of N/10 (Part 9).

(c) Acid 100 meq → 10 meq; neutralised 90 meq = 0.09 g-eq × 100 = 9 g; 9/3 = 3 g per tablet (Part 9).

Answer: (a)(iii) 31.6 (b) 0.5 L, 1.5 L (c) 3 g

PYQ drill: standards

SXC PYQ *1T-2024 Q21a, b* NEB PYQ 2076 GIE A/B Q23 NEB PYQ 2076 Set B Q23 VERY IMPORTANT **Prerequisites of a primary standard; two secondary examples** **Q.** (a) What are the prerequisites of a primary standard substance? [2] (b) Give two examples of secondary standard substances. [1] (NEB: "All standard solutions are not primary standard solutions": comment.)

(a) Easy to obtain and in a pure state; not hygroscopic, efflorescent or deliquescent; composition does not change during storage or weighing; readily soluble in water; high molecular (equivalent) mass to reduce weighing error.

(b) NaOH and $KMnO_4$ (also HCl, HNO_3).

Comment: a standard solution is any solution of exactly known concentration; only one made directly from a substance meeting the criteria above is a primary standard solution. Others (NaOH, $KMnO_4$ etc.) must be standardised first, so they are secondary.

NEB PYQ 2076 GIE Set B Q2 NEB PYQ 2073 Set D Q2 NEB PYQ 2079 GIE A Q12 NEB PYQ 2081 Set R Q20 **Why oxalic acid / Na_2CO_3 are primary standards** **Q.** Why is oxalic acid (crystal) taken as a primary standard? Why is Na_2CO_3 considered primary?

Oxalic acid crystals (126) are available in pure form, non-toxic, non-hygroscopic, non-deliquescent, of high equivalent mass (63); they can be weighed accurately and the solution is stable. Na_2CO_3 can be had pure, is stable and its solution does not change strength with time.

NEB PYQ 2075 Set B Q2 NEB PYQ 2071 Set C Q25 NEB PYQ 2080 Set J NEB PYQ 2066 Q25 NEB PYQ 2065 NEB PYQ 2061 *SXC PYQ* *NEB-2083 Q12a* VERY IMPORTANT **Distinguish primary and secondary standard (substance / solution)** **Q.** Write any two differences between primary standard and secondary standard substance / solution. [2]

See the table above: (1) primary is prepared by direct weighing, secondary is standardised against a primary; (2) primary does not change strength for a long time, secondary may change slightly; (3) examples as in the table.

NEB PYQ 2074 Supp Q2 NEB PYQ 2070 Set C Q2 NEB PYQ 2074 A / 2073 C NEB PYQ 2074 B NEB PYQ 2061 NEB PYQ 2079 D/H **Define primary standard solution; standard solution; secondary standard** **Q.** Define: primary standard solution; secondary standard solution; standard solution. [1 each]

Primary standard solution: prepared directly by dissolving a known weight of a primary standard substance in a certain volume (e.g. oxalic acid solution). **Secondary standard solution:** prepared from a secondary standard substance and standardised with the help of a primary standard solution (e.g. NaOH, HCl). **Standard solution:** a solution of precisely known concentration.

REVISION SHEET whole chapter at a glance

1. Ten lines to remember

- **Equivalent weight** E = (atomic / molecular / ionic weight) ÷ n. n = basicity (acid), acidity (base), total charge (salt), change in oxidation number (redox).
- **$KMnO_4$:** acidic medium n = 5, E = 31.6; neutral/weakly basic n = 3, E = 52.67; strongly basic n = 1, E = 158. Dilute H_2SO_4 only (HCl would be oxidised, HNO_3 is itself an oxidiser). $K_2Cr_2O_7$ acidic: n = 6, E = 49.
- **Conversions:** N = M × n; g/L = N × E = M × M_w; % w/v = g/L ÷ 10; N = % w/w × d × 10 ÷ E.
- **Law of equivalence:** g-eq of A = g-eq of B, so $N_1V_1 = N_2V_2$ (any volume unit, same on both sides). Mixing same type: $N_R = \Sigma NV \div \Sigma V$. Acid + base: $N_R = (N_1V_1 - N_2V_2) \div (V_1 + V_2)$, positive means acidic.
- **Weight from volume:** W = N × E × V(mL) ÷ 1000. Use W ÷ E for one side and NV ÷ 1000 for the other.
- **Purity:** only the pure part reacts. % purity = (pure mass ÷ impure mass) × 100.
- **Back titration:** g-eq acid taken = g-eq metal + g-eq base used on the excess. Then E = W ÷ g-eq of metal; atomic mass = E × valency.
- **$Na_2CO_3.xH_2O$:** E = 53 + 9x (n = 2 from the carbonate only). Find E = W ÷ g-eq, solve for x, round to a whole number. Indicator: methyl orange.
- **Zn / H_2:** n = PV ÷ RT (atm, L, K, R = 0.0821); mol Zn = mol H_2; % purity = (mol × 65) ÷ sample mass × 100.
- **Titration words:** titrant (T, Top, known, burette) / titrand (D, Down, determined, flask). End point (colour change) vs equivalence point (exact stoichiometric). Factor f = actual weight ÷ theoretical weight; exact N = nominal N × f.

2. Indicator and standard quick table

Titration	Equivalence pH	Indicator
Strong acid + strong base	about 7 (sharp jump 3 to 11)	methyl orange or phenolphthalein
Weak acid + strong base (e.g. oxalic + NaOH)	above 7 (8 to 10)	phenolphthalein
Strong acid + weak base (e.g. HCl + Na_2CO_3, NH_4OH)	below 7 (3 to 5)	methyl orange
Weak acid + weak base (NH_4OH + H_2CO_3)	no sharp jump	none suitable
$KMnO_4$ with oxalic acid / Fe^{2+}	redox	self-indicator (pink at end point)

	Primary standard	Secondary standard
Preparation	direct weighing of a pure substance	standardised against a primary
Stability	strength unchanged for long	strength may change
Examples	oxalic acid crystals, Na_2CO_3, Mohr's salt, $K_2Cr_2O_7$	NaOH, KOH, HCl, $KMnO_4$

Acidimetry (textbook): alkali found with standard acid. Alkalimetry: acid found with standard alkali. Exam tip: the NEB manual words these reversed, so write the definition in full.

3. Key answers recalculated in these notes

Item	Answer	Item	Answer
Molality, 93 % w/v H_2SO_4	10.43 m	Zn purity (12 g, 4.2 L, 570 mmHg, 279 K)	74.49 %
Conc. HCl 38 % (d 1.19)	12.39 M	Conc. HNO_3 needed for dilution	16.19 mL
$Na_2CO_3.xH_2O$ (0.715 g, 20 mL N/2 HCl)	x = 2	Soda crystals, 1T-2024	37.07 %; x = 10
95 % to 15 % H_2SO_4	9.39 mL acid, 92.63 g water	Lab-boy: f = 0.989	evaporate about 225 mL; oxalic 1.56 g
2076 GIE B metal	atomic mass 24	2076 GIE A equimolar mixture	157.9 mL
$CaCO_3$ / $MgCO_3$ mixture	65.35 % / 34.65 %	$KMnO_4$ standardisation (use 11.0 mL)	0.0909 N
NEB-2083 $KMnO_4$ (E = 31.5)	3.94 g/L	2079 GIE A MCQ	(c) 0.6 N
Example 10.1 (divalent metal)	E = 20, atomic mass 40	Oxalic acid crystals E	63

Where the NEB manual prints a different key (2076 GIE B: 36; 2076 GIE A: 202.6 mL), these notes use the recalculated value above.

4. PYQ checklist: tick each one when you can solve it unaided

Done	Repeated question type	Where asked	Part
[ ]	Deduce $N_1V_1 = N_2V_2$	NEB 2081 P/Q, 2080 G/H; SXC 1T-2024, PB-2026; NEB-2082	9
[ ]	End point vs equivalence point; titrant and titrand	NEB 2080 GIE B, 2081 R, 2075 GIE, 2069 A, 2064	12
[ ]	Titration error, normality factor f	NEB 2080 GIE A, 2076 C, 2074 A, 2073; SXC 1T-2024, 2T-2024, 2T-2025	13
[ ]	Primary vs secondary standard	NEB 2080 J, 2076 GIE A/B, 2075 B, 2071 C, 2073 D; SXC 1T-2024, PB-2025; NEB-2083	14
[ ]	E of $KMnO_4$ and why dilute H_2SO_4	NEB 2079 D, 2079 H, 2081 P/Q; SXC PB-2025; NEB-2082	4, 13
[ ]	Dilute with water to N/10 etc.	NEB 2077 V, 2080 GIE A, 2079 D, 2070 D, 2069 A; SXC 1T-2025, 2T-2024	9
[ ]	Mix two solutions to a target normality	NEB 2067, 2071 C, 2074 Supp, 2073 C/D; SXC PB-2025, PB-2026	9
[ ]	Divalent metal in excess acid (back titration)	NEB 2078 D, 2076 B/C, 2076 GIE B, 2074 A, 2072 C, 2070 Supp, 2065, 2079 D; SXC 2T-2025	10
[ ]	$Na_2CO_3.xH_2O$: find x	NEB 2079 GIE B, 2081 P/Q, 2064; SXC 1T-2024	11
[ ]	% purity: limestone, zinc, acid	NEB 2075 GIE (Zn), 2072 E; SXC 1T-2025	9, 11
[ ]	Conc. acid: N, M, molality, volume from % w/w and density	NEB 2080 GIE B, 2077 W, 2072 Supp, 2075 B; SXC 2T-2024, 1T-2025	5, 8
[ ]	N from M by n-factor (MCQ)	NEB 2081 P/Q, 2081 R, 2080 J, 2079 GIE A; SXC PB-2025; NEB-2083	6
[ ]	ppm / ppb numerical	see Part 7 drill	7
[ ]	Redox titration: $KMnO_4$ with oxalic acid, self-indicator	NEB-2083 Q12; SXC 1T-2024 Q21	13, 14

FORMULA SHEET every formula with its symbols named

Equivalent mass and n-factor

$$\text{Equivalent mass, } E = \frac{\text{Atomic mass / molecular mass / ionic mass}}{\text{n-factor, n}}$$

$$\text{Equivalent mass of an acid, } E_{acid} = \frac{\text{Molecular mass of the acid}}{\text{Basicity (replaceable } H^+\text{)}}$$

$$\text{Equivalent mass of a base, } E_{base} = \frac{\text{Molecular mass of the base}}{\text{Acidity (replaceable } OH^-\text{)}}$$

$$\text{Equivalent mass of a salt, } E_{salt} = \frac{\text{Molecular mass of the salt}}{\text{Total positive (or negative) charge}}$$

$$\text{Equivalent mass of an oxidising / reducing agent} = \frac{\text{Molecular mass}}{\text{Change in oxidation number per formula unit}}$$

$$\text{Equivalent mass of a metal, } E_{metal} = \frac{\text{Atomic mass of the metal}}{\text{Valency of the metal}}$$

$KMnO_4$ (M = 158): acidic n = 5, E = 31.6; neutral n = 3, E = 52.67; basic n = 1, E = 158. $K_2Cr_2O_7$ (acidic): n = 6, E = 49.

Concentration terms

$$\text{Percentage w/v} = \frac{\text{Weight of the solute in grams}}{\text{Volume of the solution in mL}} \times 100 \qquad \text{Percentage w/w} = \frac{\text{Weight of the solute in grams}}{\text{Weight of the solution in grams}} \times 100$$

$$\text{Strength in g/L} = \frac{\text{Weight of the solute in grams}}{\text{Volume of the solution in litres}}$$

$$\text{Normality of the solution, N} = \frac{\text{Weight of the solute in grams, W} \times 1000}{\text{Equivalent mass of the solute, E} \times \text{Volume of the solution in mL, V}}$$

$$\text{Molarity of the solution, M} = \frac{\text{Weight of the solute in grams, W} \times 1000}{\text{Molecular mass of the solute, } M_w \times \text{Volume of the solution in mL, V}}$$

$$\text{Molality of the solution, m} = \frac{\text{Weight of the solute in grams, W} \times 1000}{\text{Molecular mass of the solute, } M_w \times \text{Weight of the solvent in grams}}$$

$$\text{Mole fraction of the solute, } \chi_{solute} = \frac{\text{Moles of the solute}}{\text{Moles of the solute + moles of the solvent}} \qquad \chi_{solute} + \chi_{solvent} = 1$$

$$\text{Parts per million, ppm} = \frac{\text{Mass of the solute}}{\text{Mass of the solution}} \times 10^6 \qquad \text{Parts per billion, ppb} = \frac{\text{Mass of the solute}}{\text{Mass of the solution}} \times 10^9$$

$$\text{Weight of the solute in grams, W} = \frac{\text{Normality, N} \times \text{Equivalent mass, E} \times \text{Volume in mL, V}}{1000}$$

$$\text{Gram equivalents of the solute} = \frac{\text{Weight of the solute in grams, W}}{\text{Equivalent mass of the solute, E}} = \frac{\text{Normality, N} \times \text{Volume in mL, V}}{1000} \qquad \text{Milli-equivalents} = N \times V \text{ (mL)}$$

Conversions between terms

$$\text{Normality, N} = \text{Molarity, M} \times \text{n-factor, n} \qquad g/L = N \times E = M \times M_w \qquad \%\text{ w/v} = \frac{g/L}{10}$$

$$\text{Normality of the solution, N} = \frac{\%\text{ w/w} \times \text{Density, d (g/mL)} \times 10}{\text{Equivalent mass of the solute, E}} \qquad \text{Molarity of the solution, M} = \frac{\%\text{ w/w} \times \text{Density, d (g/mL)} \times 10}{\text{Molecular mass of the solute, } M_w}$$

Weight of the solution = Volume of the solution × Density, d. 1 cc = 1 mL = 1 cm^3; 1 L = 1000 mL. Molecular mass: H_2SO_4 98, HCl 36.5, HNO_3 63, NaOH 40, Na_2CO_3 106, oxalic acid 90 / crystal 126, $KMnO_4$ 158, $CaCO_3$ 100. Methyl orange 3.1 to 4.4; phenolphthalein 8.3 to 10.

Law of equivalence, dilution and mixing

Gram equivalents of A = gram equivalents of B $\Rightarrow$ $N_1V_1 = N_2V_2$ Dilution: $N_{conc}\ V_{conc} = N_{dil}\ V_{dil}$; $V_{water} = V_{dil} - V_{conc}$

$$\text{Mixing, same type: Resulting normality, } N_R = \frac{N_1V_1 + N_2V_2 + \ldots}{V_1 + V_2 + \ldots}$$

$$\text{Mixing acid and base: } N_R = \frac{N_1V_1 - N_2V_2}{V_1 + V_2}$$ (positive: acidic; negative: basic; zero: pH 7)

Purity, back titration, water of crystallisation, gas, normality factor

$$\text{Percentage purity} = \frac{\text{Mass of the pure substance}}{\text{Mass of the impure sample}} \times 100$$

Back titration: gram equivalents of acid taken = gram equivalents of metal + gram equivalents of base used for the excess

$$\text{Equivalent mass of the metal, E} = \frac{\text{Weight of the metal in grams, W}}{\text{Gram equivalents of the metal}} \qquad \text{Atomic mass} = E \times \text{valency}$$

$$E(Na_2CO_3.xH_2O) = \frac{106 + 18x}{2} = 53 + 9x$$

$$\text{Moles of gas, n} = \frac{\text{Pressure in atm, P} \times \text{Volume in litres, V}}{R \times \text{Temperature in kelvin, T}} \qquad R = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$$

$$\text{Normality factor, f} = \frac{\text{Weight of substance actually taken}}{\text{Weight of substance required}} = \frac{\text{Experimental weight}}{\text{Theoretical weight}} \qquad \text{Exact N} = \text{Nominal N} \times f$$

760 mmHg = 1 atm; T (K) = °C + 273. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$ (1 mol : 1 mol); Zn = 65.